

Table S1. ARL_1 values of the EWMA TBEA schemes when $\sigma_{0,A} = 1$ and $\sigma_{0,T} = 1$ with only the mean of T experiencing a shift based on the monitoring statistic $Z \in \{Z_1, Z_2, Z_3\}$ with both variables A and T following the gamma distribution.

	Z_1				
	EWMA				Shewhart
δ_T	$\lambda = 0.1$	$\lambda = 0.2$	$\lambda = 0.5$	$\lambda = 0.8$	$\lambda = 1$
0.95	38.9633	44.3511	59.3272	71.8214	77.7592
0.9	13.1861	14.7548	21.1327	28.7962	33.4281
0.85	6.8422	7.7153	9.4305	13.1283	15.8977
0.8	4.5037	4.4870	5.2096	6.8685	8.3641
0.75	3.3661	3.2461	3.3958	4.1037	4.8621
0.7	2.7111	2.5599	2.4823	2.7534	3.1183
0.65	2.2924	2.1307	1.9590	2.0308	2.1862
0.6	2.0022	1.8348	1.6283	1.6145	1.6703
0.55	1.7826	1.6120	1.4055	1.3627	1.3749
0.5	1.5995	1.4337	1.2516	1.2077	1.2045
	Z_2				
	EWMA				Shewhart
δ_T	$\lambda = 0.1$	$\lambda = 0.2$	$\lambda = 0.5$	$\lambda = 0.8$	$\lambda = 1$
0.95	29.6592	34.1351	47.4014	58.2581	63.3622
0.9	11.2119	11.5626	15.4874	20.543	23.5889
0.85	6.4250	6.0536	6.9190	8.8438	10.2826
0.8	4.3910	3.9427	3.9386	4.6048	5.2219
0.75	3.2869	2.8793	2.6365	2.8233	3.0639
0.7	2.6003	2.2494	1.9607	1.9727	2.0518
0.65	2.1341	1.8351	1.5679	1.5266	1.5427
0.6	1.7950	1.5439	1.3263	1.2801	1.2763
0.55	1.5354	1.3350	1.1776	1.1429	1.1361
0.5	1.3363	1.1910	1.0904	1.0691	1.0641
	Z_3				
	EWMA				Shewhart
δ_T	$\lambda = 0.1$	$\lambda = 0.2$	$\lambda = 0.5$	$\lambda = 0.8$	$\lambda = 1$
0.95	28.5995	32.4726	44.5510	55.3006	60.8680
0.9	10.7660	10.8611	13.9841	18.5382	21.5927
0.85	6.1749	5.6976	6.1834	7.7336	9.0457
0.8	4.2244	3.7245	3.5377	3.9903	4.4939
0.75	3.1661	2.7294	2.3897	2.4645	2.6328
0.7	2.5093	2.1382	1.7939	1.7515	1.7924
0.65	2.0635	1.7464	1.4493	1.3863	1.3859
0.6	1.7357	1.4689	1.2425	1.1921	1.1834

0.55	1.4806	1.2729	1.1220	1.0905	1.0835
0.5	1.2870	1.1447	1.0569	1.0402	1.0362

Table S2. ARL_1 values of the EWMA TBEA schemes when $\sigma_{0,A} = 1$ and $\sigma_{0,T} = 2$ with only the mean of T experiencing a shift based on the monitoring statistic $Z \in \{Z_1, Z_2, Z_3\}$ with both variables A and T following the gamma distribution.

	Z_1				
	EWMA				Shewhart
δ_T	$\lambda = 0.1$	$\lambda = 0.2$	$\lambda = 0.5$	$\lambda = 0.8$	$\lambda = 1$
0.95	59.7949	65.1615	79.3204	91.3778	97.2663
0.9	24.1198	26.6684	35.2412	44.7247	50.2805
0.85	12.6240	13.5768	17.7747	23.5925	27.6110
0.8	7.9811	8.2554	10.1671	13.4614	16.0971
0.75	5.7115	5.7077	6.5077	8.3080	9.9536
0.7	4.4288	4.3105	4.5687	5.5245	6.5190
0.65	3.6239	3.4585	3.4464	3.9301	4.5136
0.6	3.0799	2.8952	2.7448	2.9653	3.2957
0.55	2.6916	2.4980	2.2767	2.3521	2.5301
0.5	2.4018	2.2019	1.9475	1.9460	2.0347
	Z_2				
	EWMA				Shewhart
δ_T	$\lambda = 0.1$	$\lambda = 0.2$	$\lambda = 0.5$	$\lambda = 0.8$	$\lambda = 1$
0.95	50.5348	57.1801	72.8754	81.6557	85.2006
0.9	20.6503	22.4485	30.7195	37.001	39.9025
0.85	11.4960	11.5526	15.0794	18.6176	20.4795
0.8	7.5670	7.1645	8.5321	10.3751	11.4727
0.75	5.4716	4.9978	5.4414	6.3631	6.9825
0.7	4.1937	3.7575	3.8125	4.2541	4.5921
0.65	3.3444	2.9711	2.8697	3.0660	3.2435
0.6	2.7462	2.4364	2.2821	2.3556	2.4440
0.55	2.3079	2.0555	1.8951	1.9099	1.9503
0.5	1.9782	1.7761	1.6307	1.6199	1.6356
	Z_3				
	EWMA				Shewhart
δ_T	$\lambda = 0.1$	$\lambda = 0.2$	$\lambda = 0.5$	$\lambda = 0.8$	$\lambda = 1$
0.95	47.5831	54.3232	68.8788	77.7067	81.3182
0.9	19.0929	21.0115	28.1590	34.0631	36.8829
0.85	10.6152	10.8348	13.6768	16.8305	18.5635
0.8	7.0226	6.7587	7.7534	9.3221	10.3054
0.75	5.1146	4.7412	4.9835	5.7321	6.2675

0.7	3.9499	3.5815	3.5253	3.8633	4.1449
0.65	3.1727	2.8437	2.6792	2.8150	2.9568
0.6	2.6225	2.3409	2.1501	2.1891	2.2559
0.55	2.2172	1.9824	1.8010	1.7967	1.8245
0.5	1.9110	1.7197	1.5626	1.5417	1.5505

Table S3. ARL_1 values of the EWMA TBEA schemes when $\sigma_{0,A} = 1$ and $\sigma_{0,T} = 5$ with only the mean of T experiencing a shift based on the monitoring statistic $Z \in \{Z_1, Z_2, Z_3\}$ with both variables A and T following the gamma distribution.

	Z_1				
	EWMA				Shewhart
δ_T	$\lambda = 0.1$	$\lambda = 0.2$	$\lambda = 0.5$	$\lambda = 0.8$	$\lambda = 1$
0.95	96.9086	100.0517	108.8647	119.5153	126.8645
0.9	52.5163	54.8768	62.5691	73.5525	82.1957
0.85	31.5446	32.8524	38.0142	46.6361	54.4129
0.8	20.7240	21.3005	24.4037	30.5071	36.8009
0.75	14.6661	14.8044	16.5174	20.6073	25.4254
0.7	11.0169	10.9076	11.7435	14.3782	17.9417
0.65	8.6732	8.4295	8.7290	10.3578	12.929
0.6	7.0827	6.7700	6.7477	7.6956	9.5121
0.55	5.9513	5.6076	5.3961	5.8870	7.1432
0.5	5.1133	4.7614	4.4421	4.6266	5.4738
	Z_2				
	EWMA				Shewhart
δ_T	$\lambda = 0.1$	$\lambda = 0.2$	$\lambda = 0.5$	$\lambda = 0.8$	$\lambda = 1$
0.95	104.108	105.2551	109.3081	111.9096	112.8647
0.9	58.0625	58.9228	62.5149	65.2883	66.3401
0.85	34.5809	35.0712	37.4413	39.6860	40.5734
0.8	21.8728	22.1146	23.4865	25.1146	25.7902
0.75	14.5964	14.6923	15.4184	16.5299	17.0168
0.7	10.2069	10.2225	10.5729	11.3009	11.6395
0.65	7.4305	7.4058	7.5516	8.0126	8.2413
0.6	5.5991	5.5576	5.5983	5.8808	6.0311
0.55	4.3461	4.3008	4.2915	4.4583	4.5542
0.5	3.4615	3.4192	3.3893	3.4833	3.5423
	Z_3				
	EWMA				Shewhart
δ_T	$\lambda = 0.1$	$\lambda = 0.2$	$\lambda = 0.5$	$\lambda = 0.8$	$\lambda = 1$
0.95	91.9390	99.9046	108.0453	111.3090	112.2637
0.9	47.1163	53.4131	61.4421	64.6756	65.7294

0.85	27.0589	30.7739	36.6410	39.2034	40.0909
0.8	17.1766	19.1097	22.9186	24.7664	25.4403
0.75	11.7891	12.7074	15.0241	16.2873	16.7718
0.7	8.5699	8.9561	10.3018	11.1342	11.4703
0.65	6.4961	6.6173	7.3664	7.8984	8.1250
0.6	5.0798	5.0766	5.4723	5.8025	5.9511
0.55	4.0691	4.0136	4.2064	4.4045	4.4991
0.5	3.3234	3.2519	3.3322	3.4462	3.5043

Table S4. ARL_1 values of the EWMA TBEA schemes when $\sigma_{0,A} = 1$ and $\sigma_{0,T} = 1$ with only the mean of T experiencing a shift based on the monitoring statistic $Z \in \{Z_1, Z_2, Z_3\}$ with variables A following the normal distribution and T following the gamma distribution.

	Z_1				
	EWMA				Shewhart
δ_T	$\lambda = 0.1$	$\lambda = 0.2$	$\lambda = 0.5$	$\lambda = 0.8$	$\lambda = 1$
0.95	37.5150	42.1926	55.4252	67.0296	72.9179
0.9	12.7362	14.0197	19.3171	25.9283	30.1538
0.85	6.6610	6.8933	8.6735	11.7069	14.0881
0.8	4.4058	4.3474	4.8636	6.1718	7.4017
0.75	3.3030	3.1618	3.2140	3.7465	4.3474
0.7	2.6659	2.5012	2.3733	2.5588	2.8338
0.65	2.2568	2.0847	1.8865	1.9188	2.0317
0.6	1.9705	1.7949	1.5773	1.5479	1.5849
0.55	1.7512	1.5762	1.3695	1.3230	1.3284
0.5	1.5680	1.4030	1.2275	1.1849	1.1801
	Z_2				
	EWMA				Shewhart
δ_T	$\lambda = 0.1$	$\lambda = 0.2$	$\lambda = 0.5$	$\lambda = 0.8$	$\lambda = 1$
0.95	29.0923	33.2140	45.5697	55.8026	60.8637
0.9	11.0534	11.2904	14.753	19.3329	22.2074
0.85	6.3551	5.9470	6.6373	8.3258	9.6394
0.8	4.3518	3.8886	3.8146	4.3763	4.9254
0.75	3.2619	2.8470	2.5733	2.7154	2.9238
0.7	2.5831	2.2278	1.9249	1.9187	1.9841
0.65	2.1213	1.8194	1.5465	1.4987	1.5100
0.6	1.7847	1.5323	1.3138	1.2659	1.2609
0.55	1.5270	1.3270	1.1709	1.1362	1.1293
0.5	1.3302	1.1863	1.0872	1.0662	1.0613
	Z_3				
	EWMA				Shewhart

δ_T	$\lambda = 0.1$	$\lambda = 0.2$	$\lambda = 0.5$	$\lambda = 0.8$	$\lambda = 1$
0.95	28.0606	31.5459	42.6420	52.7939	58.0998
0.9	10.6222	10.6135	13.3289	17.4492	20.2590
0.85	6.1119	5.6031	5.9524	7.3144	8.4936
0.8	4.1887	3.6764	3.4394	3.8187	4.2622
0.75	3.1430	2.7003	2.3397	2.3869	2.5306
0.7	2.4931	2.1183	1.7651	1.7133	1.7451
0.65	2.0513	1.7316	1.4320	1.3666	1.3636
0.6	1.7257	1.4579	1.2323	1.1821	1.1729
0.55	1.4725	1.2653	1.1164	1.0856	1.0787
0.5	1.2810	1.1401	1.0542	1.0380	1.0341

Table S5. ARL_1 values of the EWMA TBEA schemes when $\sigma_{0,A} = 1$ and $\sigma_{0,T} = 2$ with only the mean of T experiencing a shift based on the monitoring statistic $Z \in \{Z_1, Z_2, Z_3\}$ with variables A following the normal distribution and T following the gamma distribution.

	Z_1				
	EWMA				Shewhart
δ_T	$\lambda = 0.1$	$\lambda = 0.2$	$\lambda = 0.5$	$\lambda = 0.8$	$\lambda = 1$
0.95	59.4090	64.3614	77.4036	89.1319	95.0766
0.9	23.8924	26.2321	34.0505	42.9968	48.4041
0.85	12.5158	13.3702	17.1489	22.5285	26.3446
0.8	7.9239	8.1497	9.8384	12.8422	15.3010
0.75	5.6771	5.6468	6.3255	7.9486	9.4635
0.7	4.4058	4.2713	4.4602	5.3114	6.2176
0.65	3.6074	3.4309	3.3768	3.7998	4.3269
0.6	3.0673	2.8741	2.6970	2.8830	3.1790
0.55	2.6815	2.4806	2.2421	2.2985	2.4565
0.5	2.3929	2.1866	1.9217	1.9102	1.9880
	Z_2				
	EWMA				Shewhart
δ_T	$\lambda = 0.1$	$\lambda = 0.2$	$\lambda = 0.5$	$\lambda = 0.8$	$\lambda = 1$
0.95	50.3392	56.9016	72.3601	81.1307	84.6452
0.9	20.5690	22.3192	30.4180	36.6357	39.5048
0.85	11.4576	11.4958	14.9299	18.4128	20.2473
0.8	7.5457	7.1362	8.4574	10.2649	11.3433
0.75	5.4585	4.9818	5.4020	6.3032	6.9104
0.7	4.1852	3.7476	3.7903	4.2207	4.5513
0.65	3.3386	2.9645	2.8564	3.0467	3.2199
0.6	2.7422	2.4319	2.2737	2.3441	2.4301
0.55	2.3050	2.0523	1.8896	1.9028	1.9419
0.5	1.9762	1.7738	1.6270	1.6155	1.6305

	Z_3				
	EWMA				Shewhart
δ_T	$\lambda = 0.1$	$\lambda = 0.2$	$\lambda = 0.5$	$\lambda = 0.8$	$\lambda = 1$
0.95	47.4308	54.1109	68.6844	77.4650	81.1217
0.9	19.0388	20.9271	28.0463	33.9160	36.7455
0.85	10.5912	10.7986	13.6212	16.7526	18.4854
0.8	7.0093	6.7404	7.7253	9.2812	10.2625
0.75	5.1063	4.7305	4.9683	5.7100	6.2437
0.7	3.9442	3.5747	3.5165	3.8508	4.1313
0.65	3.1686	2.8390	2.6737	2.8076	2.9488
0.6	2.6195	2.3376	2.1465	2.1845	2.2510
0.55	2.2149	1.9799	1.7987	1.7938	1.8215
0.5	1.9093	1.7179	1.5610	1.5398	1.5485

Table S6. ARL_1 values of the EWMA TBEA schemes when $\sigma_{0,A} = 1$ and $\sigma_{0,T} = 2$ with only the mean of T experiencing a shift based on the monitoring statistic $Z \in \{Z_1, Z_2, Z_3\}$ with variables A following the normal distribution and T following the gamma distribution.

	Z_1				
	EWMA				Shewhart
δ_T	$\lambda = 0.1$	$\lambda = 0.2$	$\lambda = 0.5$	$\lambda = 0.8$	$\lambda = 1$
0.95	96.6448	99.8923	108.2062	118.4807	125.5614
0.9	52.3538	54.7220	61.9637	72.3637	80.7013
0.85	31.4490	32.7456	37.5810	45.6479	53.0899
0.8	20.6662	21.2314	24.1158	29.7709	35.7393
0.75	14.6292	14.7594	16.3296	20.0843	24.6131
0.7	10.9919	10.8774	11.6203	14.014	17.3358
0.65	8.6551	8.4085	8.6466	10.1055	12.4834
0.6	7.0689	6.7548	6.6910	7.5202	9.1870
0.55	5.9401	5.5963	5.3556	5.7637	6.9071
0.5	5.1038	4.7529	4.4119	4.5387	5.3028
	Z_2				
	EWMA				Shewhart
δ_T	$\lambda = 0.1$	$\lambda = 0.2$	$\lambda = 0.5$	$\lambda = 0.8$	$\lambda = 1$
0.95	104.1104	105.2610	109.2972	111.9017	112.8443
0.9	58.0600	58.9216	62.5045	65.2793	66.3244
0.85	34.5786	35.0693	37.4341	39.6792	40.5630
0.8	21.8714	22.1133	23.4821	25.1102	25.7837
0.75	14.5957	14.6915	15.4158	16.5272	17.0128
0.7	10.2065	10.2221	10.5714	11.2993	11.6371
0.65	7.4303	7.4057	7.5507	8.0117	8.2399

0.6	5.5991	5.5575	5.5978	5.8802	6.0302
0.55	4.3461	4.3008	4.2913	4.4580	4.5538
0.5	3.4615	3.4192	3.3892	3.4831	3.5421
	Z₃				
	EWMA				Shewhart
δ_T	$\lambda = 0.1$	$\lambda = 0.2$	$\lambda = 0.5$	$\lambda = 0.8$	$\lambda = 1$
0.95	91.9415	99.9071	108.4839	111.3104	112.2648
0.9	47.1170	53.4139	61.6635	64.6762	65.7299
0.85	27.0591	30.7742	36.7570	39.2037	40.0911
0.8	17.1767	19.1098	22.9818	24.7665	25.4404
0.75	11.7891	12.7074	15.0597	16.2873	16.7718
0.7	8.5699	8.9561	10.3226	11.1342	11.4703
0.65	6.4961	6.6173	7.3790	7.8984	8.1250
0.6	5.0798	5.0766	5.4802	5.8025	5.9511
0.55	4.0691	4.0136	4.2114	4.4045	4.4991
0.5	3.3234	3.2519	3.3354	3.4462	3.5043

Table S7. ARL_I values of the EWMA TBFA schemes only when the mean of A experiences a shift based on the statistic $Z \in \{Z_1, Z_2, Z_3\}$ when $\sigma_{0,A} = 1$ and $\sigma_{0,T} = 1$ with both variables A and T following the gamma distribution.

Z₁											
	δ_A	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
EWMA	$\lambda = 0.1$	13.1847	4.5050	2.7120	2.0020	1.6011	1.3038	1.1112	1.0278	1.0047	1.0005
	$\lambda = 0.2$	14.7505	4.4883	2.5610	1.8355	1.4363	1.1827	1.0541	1.0109	1.0015	1.0001
	$\lambda = 0.5$	21.1162	5.2087	2.4847	1.6310	1.2535	1.0800	1.0178	1.0027	1.0003	1
	$\lambda = 0.8$	28.7717	6.8588	2.7566	1.6187	1.2092	1.0573	1.0114	1.0016	1.0001	1
Shewhart	$\lambda = 1$	33.4034	8.3450	3.1161	1.6758	1.2062	1.0522	1.0098	1.0013	1.0001	1
Z₂											
	δ_A	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
EWMA	$\lambda = 0.1$	13.6316	5.7134	3.6681	2.7588	2.2565	1.9535	1.7408	1.5440	1.3545	1.1997
	$\lambda = 0.2$	15.1909	5.3674	3.2420	2.3840	1.9327	1.643	1.4148	1.2323	1.1093	1.0429
	$\lambda = 0.5$	24.5478	6.5397	3.1761	2.0879	1.5879	1.3061	1.1399	1.0530	1.0163	1.0041
	$\lambda = 0.8$	35.848	9.5912	3.8996	2.1962	1.5326	1.2300	1.0885	1.0285	1.0075	1.0016
Shewhart	$\lambda = 1$	42.1816	12.1716	4.7435	2.4305	1.5731	1.2225	1.0781	1.0234	1.0058	1.0012
Z₃											
	δ_A	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
EWMA	$\lambda = 0.1$	13.2750	5.5486	3.5585	2.6732	2.1882	1.9398	1.7506	1.4926	1.2301	1.0683

	$\lambda = 0.2$	14.5467	5.1110	3.0887	2.2810	1.8727	1.5833	1.3045	1.1051	1.0215	1.0024
	$\lambda = 0.5$	23.2643	5.9541	2.8765	1.8969	1.4348	1.1690	1.0423	1.0059	1.0004	1
	$\lambda = 0.8$	35.1158	8.7908	3.4047	1.8858	1.3228	1.0936	1.0174	1.0017	1.0001	1
Shewhart	$\lambda = 1$	42.4946	11.5700	4.1754	2.0333	1.3199	1.0794	1.0130	1.0012	1.0001	1

Table S8. ARL_I values of the EWMA TBEA schemes only when the mean of A experiences a shift based on the statistic $Z \in \{Z_1, Z_2, Z_3\}$ when $\sigma_{0,A} = 1$ and $\sigma_{0,T} = 2$ with both variables A and T following the gamma distribution.

Z_1											
	δ_A	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
EWMA	$\lambda = 0.1$	24.0531	8.0128	4.4542	3.0958	2.4072	1.9911	1.6989	1.4714	1.2935	1.1656
	$\lambda = 0.2$	26.4713	8.2794	4.3392	2.9147	2.2165	1.8009	1.5193	1.3182	1.1793	1.0917
	$\lambda = 0.5$	34.3964	10.0793	4.6004	2.7868	1.9884	1.5650	1.3170	1.1685	1.0830	1.0375
	$\lambda = 0.8$	43.0445	13.0839	5.5136	3.0247	2.0077	1.5221	1.2686	1.1327	1.0615	1.0265
Shewhart	$\lambda = 1$	48.1395	15.4477	6.4300	3.3535	2.1108	1.5441	1.2661	1.1262	1.0567	1.0239
Z_2											
	δ_A	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
EWMA	$\lambda = 0.1$	29.3821	11.6705	7.0370	5.0361	3.9439	3.2647	2.8055	2.4763	2.2303	2.0406
	$\lambda = 0.2$	35.7568	12.3542	6.7186	4.5713	3.4913	2.8525	2.4342	2.1405	1.9230	1.7544
	$\lambda = 0.5$	57.0077	20.4823	9.5546	5.5542	3.7736	2.8437	2.2955	1.9418	1.6976	1.5203
	$\lambda = 0.8$	71.246	29.4951	14.1839	7.8526	4.9181	3.4099	2.5623	2.0494	1.7202	1.4993
Shewhart	$\lambda = 1$	77.0883	34.1912	17.1797	9.6357	5.9519	3.9996	2.8917	2.2268	1.8097	1.5392
Z_3											
	δ_A	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
EWMA	$\lambda = 0.1$	30.5837	11.4109	6.7264	4.7810	3.7395	3.0990	2.6670	2.3501	2.1173	1.9751
	$\lambda = 0.2$	40.7409	13.1887	6.7967	4.4926	3.3767	2.7346	2.3244	2.0509	1.8710	1.7310
	$\lambda = 0.5$	72.0918	26.5071	11.3103	5.9308	3.7349	2.6873	2.1155	1.7656	1.5214	1.3255
	$\lambda = 0.8$	93.9969	43.8146	20.8894	10.5592	5.8284	3.5690	2.4237	1.8010	1.4379	1.2171
Shewhart	$\lambda = 1$	102.6889	53.1285	27.8865	14.9793	8.3247	4.8509	3.0097	2.0204	1.4874	1.2079

Table S9. ARL_I values of the EWMA TBEA schemes only when the mean of A experiences a shift based on the statistic $Z \in \{Z_1, Z_2, Z_3\}$ when $\sigma_{0,A} = 1$ and $\sigma_{0,T} = 5$ with both variables A and T following the gamma distribution.

Z_1											
	δ_A	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2

EWMA	$\lambda = 0.1$	49.6218	20.0568	10.9936	7.2323	5.3206	4.2063	3.4897	2.9915	2.6223	2.3348
	$\lambda = 0.2$	50.2089	19.9450	10.6705	6.8720	4.9684	3.8678	3.1623	2.6748	2.3191	2.0493
	$\lambda = 0.5$	51.5586	19.8468	10.1031	6.2004	4.3086	3.257	2.6118	2.1866	1.8913	1.6785
	$\lambda = 0.8$	53.9363	20.7328	10.3388	6.1850	4.1992	3.1179	2.4707	2.0553	1.7745	1.5774
Shewhart	$\lambda = 1$	57.9168	22.7583	11.2911	6.6408	4.42	3.2226	2.5163	2.0705	1.7744	1.5699

Z_2

	δ_A	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
EWMA	$\lambda = 0.1$	126.329	83.2030	57.2091	41.0427	30.6597	23.7639	19.0234	15.6527	13.1796	11.3134
	$\lambda = 0.2$	130.576	88.4043	62.1585	45.3364	34.1921	26.5640	21.1851	17.2918	14.4079	12.2266
	$\lambda = 0.5$	139.337	99.8900	73.4591	55.2991	42.546	33.4116	26.7478	21.8011	18.0674	15.2043
	$\lambda = 0.8$	143.631	106.3394	80.8014	62.7930	49.7695	40.1416	32.8847	27.3202	22.9870	19.5655
Shewhart	$\lambda = 1$	145.075	108.5384	83.3654	65.4912	52.4651	42.7548	35.3709	29.6565	25.1643	21.5830

Z_3

	δ_A	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
EWMA	$\lambda = 0.1$	149.917	105.9680	71.9954	48.8275	34.1464	24.9453	19.0494	15.1395	12.4454	10.5176
	$\lambda = 0.2$	168.884	138.1791	109.3489	84.1159	63.6440	47.9641	36.3078	27.7574	21.5283	17.0009
	$\lambda = 0.5$	182.957	166.9682	151.1598	135.6493	120.6549	106.4712	93.3745	81.5141	70.893	61.4380
	$\lambda = 0.8$	187.489	175.3860	163.6844	152.3750	141.4613	130.9647	120.9251	111.3906	102.398	93.9612
Shewhart	$\lambda = 1$	188.568	177.6378	167.1925	157.2176	147.6987	138.6217	129.9727	121.7378	113.903	106.4569

Table S10. ARL_I values of the EWMA TBEA schemes only when the mean of A experiences a shift based on the statistic $Z \in \{Z_1, Z_2, Z_3\}$ when $\sigma_{0,A} = 1$ and $\sigma_{0,T} = 1$ with variables A following the normal distribution and T following the gamma distribution.

Z_1											
	δ_A	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
EWMA	$\lambda = 0.1$	12.7408	4.4094	2.6672	1.9708	1.5712	1.2808	1.1008	1.0251	1.0043	1.0005
	$\lambda = 0.2$	14.0186	4.3520	2.5033	1.7969	1.4064	1.1658	1.0485	1.0098	1.0014	1.0001
	$\lambda = 0.5$	19.2652	4.8699	2.3788	1.5811	1.2289	1.0711	1.0158	1.0024	1.0003	1
	$\lambda = 0.8$	25.7807	6.1716	2.5683	1.5533	1.1855	1.0505	1.0101	1.0014	1.0001	1
Shewhart	$\lambda = 1$	29.9389	7.3884	2.8454	1.5920	1.1807	1.0458	1.0087	1.0012	1.0001	1
Z_2											
	δ_A	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
EWMA	$\lambda = 0.1$	13.4289	5.6587	3.6403	2.7407	2.2438	1.9435	1.7300	1.5317	1.3432	1.1914
	$\lambda = 0.2$	14.7948	5.2823	3.2048	2.3617	1.9163	1.6277	1.4006	1.2217	1.1031	1.04
	$\lambda = 0.5$	23.2475	6.2592	3.0847	2.0440	1.5610	1.2886	1.1299	1.0486	1.0148	1.0037

	$\lambda = 0.8$	33.5730	8.9294	3.6905	2.1144	1.4948	1.2118	1.0804	1.0256	1.0066	1.0014
Shewhart	$\lambda = 1$	39.5779	11.2471	4.4249	2.3102	1.5242	1.2024	1.0704	1.0209	1.0051	1.0010
Z_3											
	δ_A	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
EWMA	$\lambda = 0.1$	13.0737	5.4934	3.5305	2.6544	2.1773	1.9328	1.7364	1.4731	1.2160	1.0636
	$\lambda = 0.2$	14.1569	5.0263	3.0518	2.2604	1.8569	1.5628	1.2863	1.0972	1.0201	1.0024
	$\lambda = 0.5$	22.0693	5.7005	2.7950	1.8569	1.4089	1.1554	1.0388	1.0057	1.0005	1
	$\lambda = 0.8$	33.1438	8.2008	3.2240	1.8188	1.2957	1.0850	1.0161	1.0018	1.0001	1
Shewhart	$\lambda = 1$	40.1595	10.6959	3.8812	1.9329	1.2875	1.0716	1.0121	1.0012	1.0001	1

Table S11. ARL_I values of the EWMA TBEA schemes only when the mean of A experiences a shift based on the statistic $Z \in \{Z_1, Z_2, Z_3\}$ when $\sigma_{0,A} = 1$ and $\sigma_{0,T} = 2$ with variables A following the normal distribution and T following the gamma distribution.

Z_1											
	δ_A	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
EWMA	$\lambda = 0.1$	23.8275	7.9577	4.4318	3.0831	2.3983	1.9835	1.6919	1.4652	1.2889	1.1626
	$\lambda = 0.2$	26.034	8.1765	4.3009	2.8940	2.2022	1.7897	1.5103	1.3116	1.1750	1.0894
	$\lambda = 0.5$	33.1892	9.7629	4.4965	2.7407	1.9630	1.5495	1.3074	1.1629	1.0800	1.0361
	$\lambda = 0.8$	41.2576	12.4983	5.3156	2.9460	1.9713	1.5035	1.2588	1.1276	1.0591	1.0254
Shewhart	$\lambda = 1$	46.1607	14.6959	6.1554	3.2443	2.0632	1.5219	1.2555	1.1211	1.0544	1.0228
Z_2											
	δ_A	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
EWMA	$\lambda = 0.1$	29.2585	11.6315	7.0180	5.0248	3.9363	3.2591	2.8011	2.4727	2.2273	2.038
	$\lambda = 0.2$	35.5338	12.2918	6.6939	4.5580	3.4827	2.8463	2.4293	2.1365	1.9196	1.7514
	$\lambda = 0.5$	56.4246	20.2377	9.4553	5.5079	3.7484	2.8281	2.2848	1.9339	1.6915	1.5154
	$\lambda = 0.8$	70.5468	29.0984	13.9886	7.7539	4.8647	3.3788	2.5428	2.0364	1.7111	1.4928
Shewhart	$\lambda = 1$	76.327	33.7180	16.9247	9.4975	5.874	3.9535	2.8631	2.2083	1.7974	1.5308
Z_3											
	δ_A	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
EWMA	$\lambda = 0.1$	30.4793	11.3808	6.7121	4.7725	3.7337	3.0947	2.6634	2.3469	2.1152	1.9741
	$\lambda = 0.2$	40.5616	13.1333	6.7740	4.4805	3.3691	2.7294	2.3205	2.0482	1.8690	1.7284
	$\lambda = 0.5$	71.9116	26.3908	11.2524	5.9016	3.7192	2.6782	2.1097	1.7615	1.5178	1.3222
	$\lambda = 0.8$	93.7766	43.6396	20.7775	10.4925	5.7893	3.5460	2.41	1.7926	1.4327	1.214
Shewhart	$\lambda = 1$	102.5482	52.9791	27.7693	14.8970	8.27	4.8156	2.9877	2.0072	1.4802	1.2046

Table S12. ARL_I values of the EWMA TBEA schemes only when the mean of A experiences a shift based on the statistic $Z \in \{Z_1, Z_2, Z_3\}$ when $\sigma_{0,A} = 1$ and $\sigma_{0,T} = 5$ with variables A following the normal distribution and T following the gamma distribution.

Z_1											
	δ_A	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
EWMA	$\lambda = 0.1$	49.4644	20.0021	10.9697	7.2195	7.2195	4.2008	3.4854	2.9878	2.6189	2.3317
	$\lambda = 0.2$	50.0562	19.8862	10.6460	6.8595	6.8595	3.8622	3.1579	2.6713	2.3162	2.0468
	$\lambda = 0.5$	50.9763	19.6200	10.0094	6.1555	6.1555	3.2418	2.6017	2.1794	1.8861	1.6746
	$\lambda = 0.8$	52.8289	20.3193	10.1771	6.1117	6.1117	3.0960	2.4569	2.0461	1.7682	1.5728
Shewhart	$\lambda = 1$	56.527	22.2134	11.0727	6.5416	6.5416	3.1940	2.4989	2.0592	1.7668	1.5646
Z_2											
	δ_A	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
EWMA	$\lambda = 0.1$	126.3359	83.2030	57.2082	41.0419	30.6592	23.7635	19.0232	15.6526	13.1795	11.3134
	$\lambda = 0.2$	130.5874	88.4070	62.1587	45.3360	34.1917	26.5636	21.1848	17.2916	14.4077	12.2265
	$\lambda = 0.5$	139.3253	99.8752	73.4459	55.2882	42.5374	33.4048	26.7424	21.7968	18.0639	15.2015
	$\lambda = 0.8$	143.6234	106.3268	80.7891	62.7822	49.7604	40.1340	32.8785	27.3150	22.9826	19.5618
Shewhart	$\lambda = 1$	145.0489	108.5124	83.3427	65.4722	52.4494	42.7418	35.3601	29.6475	25.1567	21.5766
Z_3											
	δ_A	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
EWMA	$\lambda = 0.1$	149.9237	105.9712	71.9971	48.8284	34.1469	24.9456	19.0494	15.1395	12.4454	10.5175
	$\lambda = 0.2$	168.8904	138.1827	109.3512	84.1176	63.6454	47.9652	36.3086	27.7580	21.5287	17.0012
	$\lambda = 0.5$	183.8021	167.7541	151.8864	136.3160	121.2623	107.0210	93.8697	81.9581	71.2894	61.7902
	$\lambda = 0.8$	187.4922	175.3885	163.6865	152.3767	141.4625	130.9655	120.9258	111.3914	102.3996	93.9621
Shewhart	$\lambda = 1$	188.5705	177.6399	167.1943	157.2192	147.7002	138.6231	129.974	121.7390	113.9048	106.4579

Table S13. ARL_1 values of the EWMA TBEA schemes for different statistics Z_i when the means of both A and T experience a shift based on $\lambda = 0.1$ when $\sigma_{0,A} = 1$ and $\sigma_{0,T} = 1$ with both variables A and T following the gamma distribution.

Z_1										
$\delta_T \backslash \delta_A$	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
0.95	6.8428	3.3671	2.2926	1.7826	1.4417	1.1923	1.0585	1.0121	1.0017	1.0002
0.9	4.4045	2.7118	2.0020	1.6009	1.3035	1.1112	1.0280	1.0048	1.0006	1
0.85	3.3667	2.2926	1.7826	1.4414	1.1922	1.0586	1.0122	1.0018	1.0002	1
0.8	2.7115	2.0021	1.6007	1.3032	1.1113	1.0282	1.0049	1.0006	1.0001	1
0.75	2.2925	1.7826	1.4410	1.1920	1.0588	1.0124	1.0018	1.0002	1	1
0.7	2.0021	1.6004	1.3027	1.1113	1.0285	1.0051	1.0006	1.0001	1	1
0.65	1.7826	1.4404	1.1917	1.0591	1.0127	1.0019	1.0002	1	1	1
0.6	1.6001	1.3020	1.1113	1.0289	1.0053	1.0007	1.0001	1	1	1
0.55	1.4396	1.1912	1.0595	1.0131	1.0021	1.0002	1	1	1	1
0.5	1.3011	1.1114	1.0294	1.0056	1.0008	1.0001	1	1	1	1
Z_2										
$\delta_T \backslash \delta_A$	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
0.95	7.4811	4.2038	2.9869	2.3673	2.0057	1.7663	1.5621	1.3698	1.2110	1.1033
0.9	5.0264	3.2952	2.5085	2.0723	1.7974	1.5815	1.3854	1.2227	1.1107	1.0473
0.85	3.7320	2.6924	2.1571	1.8362	1.6030	1.4015	1.2348	1.1184	1.0514	1.0193
0.8	2.9399	2.2663	1.8852	1.6277	1.4183	1.2472	1.1265	1.0558	1.0213	1.0071
0.75	2.4089	1.9478	1.6570	1.4363	1.2600	1.1349	1.0605	1.0236	1.008	1.0024
0.7	2.0285	1.6929	1.4559	1.2733	1.1436	1.0655	1.0260	1.0091	1.0028	1.0008
0.65	1.7378	1.4781	1.2872	1.1527	1.0708	1.0287	1.0102	1.0032	1.0009	1.0002
0.6	1.5041	1.3020	1.1622	1.0764	1.0316	1.0116	1.0038	1.0011	1.0003	1.0001
0.55	1.3182	1.1721	1.0824	1.0348	1.0130	1.0044	1.0013	1.0004	1.0001	1
0.5	1.1825	1.0886	1.0382	1.0147	1.0051	1.0016	1.0005	1.0001	1	1
Z_3										
$\delta_T \backslash \delta_A$	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
0.95	7.5493	4.2546	3.0187	2.3850	2.0352	1.8334	1.6145	1.3524	1.1405	1.0353
0.9	5.1396	3.4149	2.6055	2.1553	1.9013	1.7006	1.4614	1.2285	1.0775	1.0165
0.85	3.8375	2.8296	2.2854	1.9694	1.7584	1.5425	1.3131	1.1333	1.0386	1.0071
0.8	3.0298	2.4032	2.0345	1.7979	1.5935	1.3791	1.1902	1.0697	1.0174	1.0028
0.75	2.4854	2.0820	1.8214	1.6179	1.4196	1.2363	1.1032	1.0328	1.0071	1.0010
0.7	2.0957	1.8235	1.6186	1.4340	1.2639	1.1309	1.0501	1.0140	1.0027	1.0003
0.65	1.7963	1.5958	1.4244	1.2711	1.1472	1.0648	1.0221	1.0055	1.0010	1.0001
0.6	1.5490	1.3937	1.2592	1.1500	1.0733	1.0290	1.0089	1.0020	1.0003	1
0.55	1.3459	1.2323	1.1406	1.0743	1.0331	1.0120	1.0034	1.0007	1.0001	1
0.5	1.1956	1.1225	1.0687	1.0336	1.0139	1.0047	1.0013	1.0003	1	1

Table S14. ARL_1 values of the EWMA TBEA schemes for different statistics Z_i when the means of both A and T experience a shift based on $\lambda = 0.1$ when $\sigma_{0,A} = 1$ and $\sigma_{0,T} = 2$ with both variables A and T following the gamma distribution.

Z_1										
$\delta_T \backslash \delta_A$	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
0.95	12.6367	5.7365	3.6426	2.7015	2.1771	1.8343	1.5780	1.3755	1.2229	1.1197
0.9	7.9985	4.4475	3.0931	2.4065	1.9907	1.6978	1.4697	1.2922	1.1653	1.0846
0.85	5.7254	3.6378	2.6998	2.1767	1.8335	1.5763	1.3738	1.2220	1.1198	1.0588
0.8	4.4393	3.0898	2.4056	1.9902	1.6964	1.4676	1.2906	1.1648	1.0852	1.0403
0.75	3.6318	2.6978	2.1763	1.8326	1.5741	1.3715	1.2208	1.1199	1.0596	1.0273
0.7	3.0855	2.4046	1.9897	1.6945	1.4648	1.2885	1.1643	1.0858	1.0413	1.0185
0.65	2.6951	2.1758	1.8313	1.5713	1.3685	1.2193	1.1200	1.0606	1.0285	1.0126
0.6	2.4033	1.9891	1.6921	1.4611	1.2857	1.1634	1.0865	1.0427	1.0198	1.0087
0.55	2.1754	1.8297	1.5673	1.3643	1.2170	1.1200	1.0619	1.0301	1.0138	1.0061
0.5	1.9886	1.6886	1.4558	1.2816	1.1621	1.0873	1.0444	1.0214	1.0099	1.0044
Z_2										
$\delta_T \backslash \delta_A$	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
0.95	14.9275	8.0122	5.4410	4.1389	3.3644	2.8561	2.4995	2.2371	2.0367	1.8786
0.9	9.4005	5.9583	4.3735	3.4799	2.9132	2.5250	2.2443	2.0325	1.8669	1.7326
0.85	6.6388	4.6604	3.6149	2.9779	2.5532	2.2520	2.0279	1.8546	1.7155	1.5994
0.8	5.0174	3.7742	3.0515	2.5845	2.2602	2.0230	1.8418	1.6979	1.5794	1.4790
0.75	3.9643	3.1358	2.6191	2.2691	2.0177	1.8283	1.6799	1.5593	1.4586	1.3728
0.7	3.2326	2.6574	2.2786	2.0118	1.8142	1.6614	1.5391	1.4383	1.3539	1.2827
0.65	2.6998	2.2885	2.0054	1.7993	1.6425	1.5187	1.4184	1.3355	1.2666	1.2094
0.6	2.2984	1.9980	1.7835	1.6230	1.4983	1.3987	1.3177	1.2512	1.1966	1.1522
0.55	1.9892	1.7665	1.6027	1.4776	1.3792	1.3003	1.2364	1.1846	1.1428	1.1094
0.5	1.7478	1.5816	1.4567	1.3599	1.2835	1.2224	1.1734	1.1341	1.1029	1.0782
Z_3										
$\delta_T \backslash \delta_A$	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
0.95	15.1583	8.0648	5.4407	4.1249	3.3478	2.8403	2.5826	2.2167	2.0295	1.9130
0.9	9.4849	6.1213	4.5156	3.5969	3.0105	2.6072	2.3133	2.0956	1.9449	1.8410
0.85	6.7038	4.8591	3.8191	3.1632	2.7174	2.3972	2.1587	1.9827	1.8570	1.7568
0.8	5.0826	3.9770	3.2772	2.8015	2.4611	2.2079	2.0163	1.8726	1.7617	1.6616
0.75	4.0298	3.3281	2.8447	2.4960	2.2356	2.0363	1.8821	1.7613	1.6593	1.5601
0.7	3.2957	2.8328	2.4928	2.2353	2.0358	1.8787	1.7529	1.6482	1.5533	1.4588
0.65	2.7580	2.4445	2.2020	2.0108	1.8575	1.7326	1.6281	1.5363	1.4497	1.3639
0.6	2.3505	2.1340	1.9593	1.8164	1.6978	1.5976	1.5102	1.4304	1.3543	1.2804
0.55	2.0344	1.8827	1.7557	1.6482	1.5561	1.4755	1.4029	1.3355	1.2715	1.2108
0.5	1.7860	1.6786	1.5856	1.5046	1.4331	1.3688	1.3099	1.2550	1.2035	1.1557

Table S15. ARL_1 values of the EWMA TBEA schemes for different statistics Z_i when the means of both A and T experience a shift based on $\lambda = 0.1$ when $\sigma_{0,A} = 1$ and $\sigma_{0,T} = 5$ with both variables A and T following the gamma distribution.

Z_1										
$\delta_T \backslash \delta_A$	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
0.95	30.4864	14.4569	8.7486	6.1188	4.6834	3.8040	3.2160	2.7927	2.4693	2.2117
0.9	20.3406	10.9904	7.1973	5.2887	4.1832	3.4764	2.9858	2.6202	2.3329	2.0993
0.85	14.5423	8.7204	6.0815	4.6524	3.7831	3.2057	2.7896	2.4684	2.2093	1.9958
0.8	10.9963	7.1593	5.2509	4.1535	3.4581	2.9791	2.6197	2.3324	2.0960	1.8995
0.75	8.6940	6.0395	4.6147	3.7549	3.1908	2.7868	2.4701	2.2087	1.9907	1.8094
0.7	7.1199	5.2072	4.1157	3.4317	2.9684	2.6214	2.3358	2.0942	1.8920	1.7248
0.65	5.9946	4.5697	3.7171	3.1668	2.7815	2.4765	2.2129	1.9867	1.7988	1.6453
0.6	5.1593	4.0691	3.3939	2.9478	2.6227	2.3470	2.0979	1.8845	1.7104	1.5706
0.55	4.5188	3.6682	3.1293	2.7657	2.4860	2.2285	1.9884	1.7866	1.6265	1.5007
0.5	4.0142	3.3422	2.9111	2.6135	2.3663	2.1173	1.8823	1.6923	1.5471	1.4354
Z_2										
$\delta_T \backslash \delta_A$	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
0.95	69.5796	48.5223	35.2571	26.6289	20.8326	16.8094	13.9258	11.7959	10.1796	8.9234
0.9	41.0135	30.1643	23.0300	18.1867	14.7937	12.3425	10.5199	9.1289	8.0423	7.1762
0.85	25.7202	19.8428	15.8164	12.9714	10.9005	9.3506	8.1609	7.2269	6.4790	5.8697
0.8	17.0416	13.7079	11.3347	9.5953	8.2853	7.2741	6.4763	5.8346	5.3096	4.8738
0.75	11.8443	9.8737	8.4205	7.3199	6.4659	5.7889	5.2419	4.7925	4.4181	4.1021
0.7	8.5765	7.3687	6.4494	5.7327	5.1620	4.6990	4.3171	3.9977	3.7272	3.4955
0.65	6.4312	5.6679	5.0704	4.5928	4.2037	3.8816	3.6113	3.3815	3.1841	3.0129
0.6	4.9690	4.4744	4.0777	3.7534	3.4841	3.2573	3.0639	2.8973	2.7522	2.6250
0.55	3.9397	3.6130	3.3454	3.1224	2.9340	2.7729	2.6336	2.5121	2.4052	2.3105
0.5	3.1951	2.9765	2.7940	2.6394	2.5068	2.3919	2.2914	2.2027	2.1239	2.0534
Z_3										
$\delta_T \backslash \delta_A$	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
0.95	70.6361	53.7293	39.0309	29.2808	22.5435	17.8695	14.5656	12.1736	10.3975	9.0457
0.9	37.8305	30.1104	24.0601	19.4853	16.0597	13.4785	11.5085	9.9819	8.7801	7.8191
0.85	22.8038	19.2136	16.2823	13.9285	12.0440	10.5288	9.3013	8.2982	7.4708	6.7821
0.8	15.0980	13.2942	11.7560	10.4552	9.3570	8.4281	7.6397	6.9676	6.3919	5.8965
0.75	10.7063	9.7371	8.8781	8.1205	7.4537	6.8669	6.3500	5.8942	5.4915	5.1350
0.7	7.9737	7.4248	6.9224	6.4645	6.0480	5.6698	5.3266	5.0154	4.7333	4.4775
0.65	6.1533	5.8300	5.5265	5.2426	4.9776	4.7308	4.5014	4.2885	4.0912	3.9087
0.6	4.8764	4.6808	4.4935	4.3145	4.1440	3.9820	3.8283	3.6829	3.5457	3.4165
0.55	3.9457	3.8253	3.7082	3.5944	3.4842	3.3776	3.2750	3.1762	3.0816	2.9910
0.5	3.2477	3.1729	3.0992	3.0267	2.9555	2.8857	2.8176	2.7511	2.6865	2.6239

Table S16. ARL_1 values of the EWMA TBEA schemes for different statistics Z_i when the means of both A and T experience a shift based on $\lambda = 0.2$ when $\sigma_{0,A} = 1$ and $\sigma_{0,T} = 1$ with both variables A and T following the gamma distribution.

Z_1										
$\delta_T \backslash \delta_A$	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
0.95	7.1754	3.2472	2.1314	1.6134	1.2931	1.1042	1.0256	1.0043	1.0005	1
0.9	4.4877	2.5607	1.8354	1.4361	1.1826	1.0542	1.0110	1.0015	1.0001	1
0.85	3.2467	2.1312	1.6132	1.2929	1.1043	1.0257	1.0043	1.0005	1	1
0.8	2.5604	1.8353	1.4358	1.1824	1.0544	1.0112	1.0016	1.0002	1	1
0.75	2.1310	1.6130	1.2925	1.1043	1.0260	1.0045	1.0005	1	1	1
0.7	1.8351	1.4353	1.1822	1.0546	1.0114	1.0017	1.0002	1	1	1
0.65	1.6126	1.2920	1.1044	1.0263	1.0047	1.0006	1.0001	1	1	1
0.6	1.4347	1.1819	1.0550	1.0117	1.0018	1.0002	1	1	1	1
0.55	1.2913	1.1045	1.0267	1.0049	1.0007	1.0001	1	1	1	1
0.5	1.1815	1.0554	1.0122	1.0020	1.0002	1	1	1	1	1
Z_2										
$\delta_T \backslash \delta_A$	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
0.95	7.3277	3.7713	2.5948	2.0311	1.6966	1.4521	1.2594	1.1259	1.0511	1.0173
0.9	4.6041	2.8850	2.1582	1.7609	1.4929	1.2884	1.1442	1.0605	1.0212	1.0063
0.85	3.3029	2.3263	1.8403	1.5386	1.3196	1.1642	1.0711	1.0259	1.0079	1.0021
0.8	2.5552	1.9412	1.5917	1.3535	1.1861	1.0832	1.0313	1.0100	1.0027	1.0006
0.75	2.0727	1.6554	1.3911	1.2100	1.0967	1.0377	1.0125	1.0035	1.0009	1.0002
0.7	1.7343	1.4337	1.2361	1.1118	1.0450	1.0155	1.0046	1.0012	1.0003	1.0001
0.65	1.4836	1.2650	1.1285	1.0535	1.0190	1.0058	1.0016	1.0004	1.0001	1
0.6	1.2976	1.1471	1.0631	1.0233	1.0075	1.0021	1.0005	1.0001	1	1
0.55	1.1678	1.0740	1.0283	1.0094	1.0028	1.0007	1.0002	1	1	1
0.5	1.0864	1.0342	1.0119	1.0037	1.0010	1.0003	1.0001	1	1	1
Z_3										
$\delta_T \backslash \delta_A$	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
0.95	7.3028	3.7665	2.5887	2.0349	1.7108	1.4346	1.1955	1.0571	1.0099	1.0009
0.9	4.6537	2.9565	2.2219	1.8263	1.5454	1.2950	1.1134	1.0281	1.0041	1.0003
0.85	3.3571	2.4209	1.9385	1.6336	1.3838	1.1804	1.0593	1.0126	1.0016	1.0001
0.8	2.6036	2.0404	1.7022	1.4511	1.2432	1.0988	1.0279	1.0051	1.0006	1
0.75	2.1140	1.7492	1.4943	1.2906	1.1376	1.0486	1.0119	1.0019	1.0002	1
0.7	1.7666	1.5152	1.3171	1.1671	1.0696	1.0216	1.0047	1.0007	1.0001	1
0.65	1.5041	1.3218	1.1821	1.0857	1.0317	1.0088	1.0017	1.0002	1	1
0.6	1.3059	1.1818	1.0935	1.0396	1.0132	1.0033	1.0006	1.0001	1	1
0.55	1.1680	1.0922	1.0434	1.0168	1.0052	1.0012	1.0002	1	1	1
0.5	1.0836	1.0426	1.0186	1.0067	1.0019	1.0004	1	1	1	1

Table S17. ARL_1 values of the EWMA TBEA schemes for different statistics Z_i when the means of both A and T experience a shift based on $\lambda = 0.2$ when $\sigma_{0,A} = 1$ and $\sigma_{0,T} = 2$ with both variables A and T following the gamma distribution.

Z_1										
$\delta_T \backslash \delta_A$	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
0.95	13.5614	5.7339	3.4802	2.5128	1.9853	1.6472	1.4095	1.2412	1.1299	1.0635
0.9	8.2684	4.3316	2.9113	2.2147	1.7993	1.5175	1.3169	1.1789	1.0921	1.0433
0.85	5.7222	3.4745	2.5102	1.9835	1.6453	1.4076	1.2403	1.1299	1.0642	1.0291
0.8	4.3222	2.9071	2.2125	1.7972	1.5153	1.3152	1.1784	1.0926	1.0441	1.0194
0.75	3.4675	2.5071	1.9813	1.6427	1.4053	1.2390	1.1300	1.0650	1.0300	1.0129
0.7	2.9019	2.2098	1.7946	1.5124	1.3131	1.1778	1.0932	1.0452	1.0204	1.0086
0.65	2.5031	1.9784	1.6394	1.4022	1.2373	1.1300	1.0660	1.0312	1.0139	1.0058
0.6	2.2063	1.7910	1.5084	1.3101	1.1768	1.0938	1.0465	1.0217	1.0095	1.0040
0.55	1.9746	1.6348	1.3978	1.2349	1.1298	1.0672	1.0328	1.0152	1.0067	1.0028
0.5	1.7860	1.5026	1.3058	1.1752	1.0945	1.0482	1.0234	1.0109	1.0049	1.0021
Z_2										
$\delta_T \backslash \delta_A$	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
0.95	16.216	7.7947	4.9859	3.6827	2.9489	2.4837	2.1644	1.9320	1.7545	1.6130
0.9	9.3207	5.5188	3.9142	3.0612	2.5400	2.1912	1.9422	1.7551	1.6082	1.4886
0.85	6.2200	4.1980	3.1928	2.6041	2.2214	1.9539	1.7563	1.6036	1.4813	1.3808
0.8	4.5513	3.3486	2.6776	2.2554	1.9671	1.7582	1.5994	1.4741	1.3728	1.2898
0.75	3.5343	2.7620	2.2935	1.9819	1.7606	1.5954	1.4671	1.3649	1.2823	1.2154
0.7	2.8591	2.3362	1.9983	1.7636	1.5915	1.4602	1.3572	1.2750	1.2092	1.1569
0.65	2.3838	2.0163	1.7670	1.5878	1.4534	1.3495	1.2678	1.2032	1.1522	1.1124
0.6	2.0355	1.7705	1.5839	1.4464	1.3419	1.2608	1.1973	1.1477	1.1092	1.0797
0.55	1.7738	1.5797	1.4393	1.3342	1.2538	1.1916	1.1434	1.1062	1.0778	1.0563
0.5	1.5747	1.4316	1.3263	1.2468	1.1860	1.1393	1.1034	1.0761	1.0555	1.0400
Z_3										
$\delta_T \backslash \delta_A$	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
0.95	17.6349	8.3774	5.2207	3.7794	2.9845	2.4912	2.1626	1.9389	1.7806	1.6397
0.9	9.9055	5.9556	4.1884	3.2376	2.6593	2.2783	2.0152	1.8290	1.6838	1.5433
0.85	6.5318	4.5387	3.4636	2.8127	2.3858	2.0901	1.8780	1.7188	1.5826	1.4472
0.8	4.7460	3.6207	2.9287	2.4712	2.1523	1.9213	1.7479	1.6084	1.4817	1.3568
0.75	3.6675	2.9825	2.5192	2.1913	1.9506	1.7682	1.6240	1.5009	1.3861	1.2763
0.7	2.9553	2.5161	2.1970	1.9581	1.7745	1.6289	1.5078	1.4004	1.3004	1.2083
0.65	2.4554	2.1632	1.9385	1.7621	1.6206	1.5035	1.4024	1.3111	1.2275	1.1535
0.6	2.0895	1.8897	1.7288	1.5972	1.4875	1.3935	1.3105	1.2357	1.1686	1.1113
0.55	1.8145	1.6752	1.5585	1.4598	1.3749	1.3005	1.2341	1.1748	1.1229	1.0799
0.5	1.6051	1.5066	1.4216	1.3476	1.2826	1.2249	1.1734	1.1279	1.0889	1.0572

Table S18. ARL_1 values of the EWMA TBEA schemes for different statistics Z_i when the means of both A and T experience a shift based on $\lambda = 0.2$ when $\sigma_{0,A} = 1$ and $\sigma_{0,T} = 5$ with both variables A and T following the gamma distribution.

Z_1										
$\delta_T \backslash \delta_A$	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
0.95	30.9455	14.2571	8.4080	5.7658	4.3428	3.4739	2.8932	2.4793	2.1706	1.9329
0.9	20.5060	10.7178	6.8503	4.9465	3.8534	3.1520	2.6654	2.3093	2.0390	1.8286
0.85	14.4791	8.4038	5.7378	4.3229	3.4619	2.8839	2.4696	2.1598	1.9215	1.7347
0.8	10.7891	6.8229	4.9175	3.8367	3.1425	2.6566	2.2987	2.0268	1.8158	1.6499
0.75	8.4061	5.7005	4.2959	3.4493	2.8765	2.4605	2.1476	1.9074	1.7204	1.5732
0.7	6.7927	4.8768	3.8136	3.1344	2.6508	2.2884	2.0124	1.7997	1.6340	1.5039
0.65	5.6545	4.2553	3.4318	2.8733	2.4553	2.1351	1.8902	1.7019	1.5557	1.4414
0.6	4.8228	3.7755	3.1239	2.6524	2.2827	1.9965	1.7789	1.6129	1.4848	1.3851
0.55	4.1970	3.3980	2.8714	2.4614	2.1271	1.8698	1.6772	1.5319	1.4207	1.3345
0.5	3.7149	3.0966	2.6608	2.2924	1.9845	1.7529	1.5838	1.4581	1.3627	1.2892
Z_2										
$\delta_T \backslash \delta_A$	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
0.95	72.0592	51.2587	37.8219	28.8489	22.6572	18.2552	15.0423	12.6427	10.8131	9.3919
0.9	42.3511	31.5850	24.3489	19.3213	15.7210	13.0736	11.0813	9.5510	8.3536	7.4009
0.85	26.4092	20.5609	16.4754	13.5321	11.3536	9.7030	8.4264	7.4209	6.6158	5.9614
0.8	17.3735	14.0494	11.6434	9.8529	8.4883	7.4265	6.5851	5.9073	5.3532	4.8943
0.75	11.9827	10.0155	8.5455	7.4197	6.5391	5.8375	5.2692	4.8022	4.4135	4.0860
0.7	8.6132	7.4076	6.4809	5.7529	5.1701	4.6958	4.3041	3.9766	3.6994	3.4625
0.65	6.4179	5.6569	5.0572	4.5752	4.1813	3.8547	3.5803	3.3472	3.1472	2.9738
0.6	4.9343	4.4422	4.0456	3.7205	3.4498	3.2216	3.0270	2.8594	2.7136	2.5858
0.55	3.8990	3.5746	3.3081	3.0856	2.8974	2.7363	2.5971	2.4756	2.3688	2.2743
0.5	3.1559	2.9393	2.7581	2.6045	2.4726	2.3583	2.2583	2.1701	2.0917	2.0217
Z_3										
$\delta_T \backslash \delta_A$	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
0.95	84.8958	70.4603	57.2370	45.8102	36.4520	29.0547	23.3061	18.8721	15.4636	12.8440
0.9	46.0604	39.1205	32.8390	27.3951	22.8397	19.1068	16.0778	13.6309	11.6577	10.0662
0.85	27.0897	23.6440	20.5238	17.7812	15.4225	13.4191	11.7281	10.3050	9.1092	8.1044
0.8	17.2067	15.4264	13.7987	12.3391	11.0477	9.9141	8.9236	8.0603	7.3092	6.6560
0.75	11.6886	10.7287	9.8382	9.0221	8.2808	7.6111	7.0084	6.4674	5.9827	5.5491
0.7	8.3901	7.8510	7.3426	6.8672	6.4252	6.0160	5.6386	5.2913	4.9725	4.6804
0.65	6.2918	5.9781	5.6776	5.3914	5.1201	4.8638	4.6225	4.3958	4.1834	3.9849
0.6	4.8839	4.6960	4.5136	4.3371	4.1671	4.0037	3.8472	3.6977	3.5552	3.4198
0.55	3.8969	3.7819	3.6690	3.5584	3.4504	3.3451	3.2429	3.1437	3.0479	2.9555
0.5	3.1802	3.1089	3.0381	2.9681	2.8990	2.8309	2.7639	2.6982	2.6340	2.5712

Table S19. ARL_I values of the EWMA TBEA schemes for different statistics Z_i when the means of both A and T experience a shift based on $\lambda = 0.5$ when $\sigma_{0,A} = 1$ and $\sigma_{0,T} = 1$ with both variables A and T following the gamma distribution.

Z_1										
$\delta_T \backslash \delta_A$	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
0.95	9.4250	3.3972	1.9613	1.4079	1.1481	1.0396	1.0073	1.0009	1.0001	1
0.9	5.2090	2.4841	1.6306	1.2533	1.0801	1.0180	1.0028	1.0003	1	1
0.85	3.3966	1.9608	1.4079	1.1481	1.0398	1.0074	1.0010	1.0001	1	1
0.8	5.4834	1.6301	1.2531	1.0802	1.0181	1.0028	1.0003	1	1	1
0.75	1.9600	1.4071	1.0400	1.0400	1.0076	1.0010	1.0001	1	1	1
0.7	1.6293	1.2528	1.0804	1.0184	1.0030	1.0003	1	1	1	1
0.65	1.4064	1.1479	1.0400	1.0078	1.0011	1.0001	1	1	1	1
0.6	1.2523	1.0806	1.0187	1.0031	1.0004	1	1	1	1	1
0.55	1.1478	1.0408	1.0082	1.0012	1.0001	1	1	1	1	1
0.5	1.0809	1.0193	1.0034	1.0004	1	1	1	1	1	1
Z_2										
$\delta_T \backslash \delta_A$	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
0.95	9.6096	3.8839	2.3347	1.6998	1.3663	1.1731	1.0689	1.0225	1.0060	1.0013
0.9	5.0115	2.6784	1.8421	1.4381	1.2122	1.0884	1.0304	1.0085	1.0020	1.0004
0.85	3.1776	2.0284	1.5256	1.2586	1.1120	1.0406	1.0121	1.003	1.0006	1.0001
0.8	2.2811	1.6349	1.3139	1.1405	1.0534	1.0168	1.0044	1.0009	1.0002	1
0.75	1.7752	1.3809	1.1746	1.0693	1.0230	1.0063	1.0015	1.0003	1	1
0.7	1.4635	1.2156	1.0889	1.0311	1.0091	1.0022	1.0005	1.0001	1	1
0.65	1.2653	1.1129	1.0413	1.0128	1.0033	1.0007	1.0001	1	1	1
0.6	1.1421	1.0543	1.0177	1.0049	1.0012	1.0002	1	1	1	1
0.55	1.0704	1.0241	1.0071	1.0018	1.0004	1.0001	1	1	1	1
0.5	1.0325	1.0101	1.0027	1.0006	1.0001	1	1	1	1	1
Z_3										
$\delta_T \backslash \delta_A$	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
0.95	9.3693	3.7680	2.2502	1.6257	1.2869	1.0975	1.0207	1.0025	1.0001	1
0.9	4.9405	2.6803	1.8376	1.4172	1.1742	1.0510	1.0092	1.0009	1	1
0.85	3.1447	2.0590	1.5481	1.2595	1.0960	1.0241	1.0037	1.0003	1	1
0.8	2.2586	1.6668	1.3402	1.1475	1.0477	1.0104	1.0014	1.0001	1	1
0.75	1.7551	1.4050	1.1953	1.0759	1.0215	1.0041	1.0005	1	1	1
0.7	1.4437	1.2302	1.1019	1.0353	1.0088	1.0015	1.0002	1	1	1
0.65	1.2466	1.1200	1.0482	1.0149	1.0033	1.0005	1	1	1	1
0.6	1.1268	1.0570	1.0208	1.0059	1.0012	1.0002	1	1	1	1
0.55	1.0599	1.0249	1.0084	1.0022	1.0004	1.0001	1	1	1	1
0.5	1.0263	1.0102	1.0032	1.0008	1.0001	1	1	1	1	1

Table S20. ARL_I values of the EWMA TBEA schemes for different statistics Z_i when the means of both A and T experience a shift based on $\lambda = 0.5$ when $\sigma_{0,A} = 1$ and $\sigma_{0,T} = 2$ with both variables A and T following the gamma distribution.

Z_1										
$\delta_T \backslash \delta_A$	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
0.95	17.5479	6.5079	3.4828	2.3154	1.7452	1.4243	1.2328	1.1196	1.0568	1.0247
0.9	10.1166	4.5915	2.7795	1.9838	1.5622	1.3157	1.1683	1.0836	1.0383	1.0162
0.85	6.5069	3.4730	2.3090	1.7412	1.4221	1.2320	1.1199	1.0575	1.0255	1.0105
0.8	4.5810	2.7704	1.9780	1.5587	1.3140	1.1681	1.0842	1.0392	1.0169	1.0068
0.75	3.4611	2.3009	1.7361	1.4191	1.2309	1.1202	1.0584	1.0265	1.0112	1.0044
0.7	2.7592	1.9707	1.5541	1.3117	1.1677	1.0849	1.0403	1.0179	1.0075	1.0029
0.65	2.2905	1.7293	1.4152	1.2294	1.1205	1.0596	1.0277	1.0121	1.0050	1.0020
0.6	1.9609	1.5479	1.3086	1.1671	1.0858	1.0417	1.0191	1.0083	1.0035	1.0014
0.55	1.7200	1.4097	1.2272	1.1207	1.0610	1.0293	1.0134	1.0059	1.0025	1.0010
0.5	1.5391	1.3040	1.1660	1.0869	1.0435	1.0208	1.0096	1.0043	1.0018	1.0008
Z_2										
$\delta_T \backslash \delta_A$	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
0.95	25.0099	11.018	6.1074	4.0145	2.9600	2.3554	1.9736	1.7146	1.5294	1.3919
0.9	12.8357	6.7720	4.2941	3.0913	2.4216	2.0083	1.7330	1.5391	1.3970	1.2904
0.85	7.5718	4.6197	3.2397	2.4947	2.0459	1.7527	1.5494	1.4024	1.2933	1.2114
0.8	4.9993	3.4077	2.5755	2.0867	1.7737	1.5603	1.4080	1.2963	1.2132	1.1513
0.75	3.5978	2.6646	2.1309	1.7961	1.5717	1.4139	1.2995	1.2151	1.1526	1.1066
0.7	2.7626	2.1783	1.8198	1.5836	1.4199	1.3027	1.2170	1.1540	1.1078	1.0742
0.65	2.2288	1.8444	1.5958	1.4260	1.3058	1.2189	1.1554	1.1090	1.0754	1.0514
0.6	1.8697	1.6080	1.4320	1.3089	1.2207	1.1568	1.1103	1.0767	1.0526	1.0356
0.55	1.6198	1.4376	1.3118	1.2225	1.1582	1.1116	1.0780	1.0539	1.0368	1.0249
0.5	1.4425	1.3141	1.2240	1.1595	1.1130	1.0795	1.0554	1.0382	1.0261	1.0177
Z_3										
$\delta_T \backslash \delta_A$	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
0.95	29.3506	13.4224	7.0826	4.3610	2.0483	2.3374	1.9116	1.6304	1.4216	1.2526
0.9	14.2887	7.9029	4.9055	3.3940	2.5611	2.0617	1.7379	1.5093	1.3322	1.1909
0.85	8.1393	5.2322	3.6578	2.7539	2.2003	1.8395	1.5891	1.4025	1.2551	1.1407
0.8	5.2553	3.7784	2.8785	2.3058	1.9242	1.6577	1.4620	1.3107	1.1910	1.1015
0.75	3.7184	2.9079	2.3584	1.9790	1.7084	1.5085	1.3553	1.2343	1.1399	1.0719
0.7	2.8359	2.3467	1.9938	1.7339	1.5380	1.3869	1.2676	1.1730	1.1006	1.0503
0.65	2.2727	1.9650	1.7295	1.5471	1.4037	1.2894	1.1978	1.1255	1.0713	1.0349
0.6	1.8971	1.6956	1.5343	1.4042	1.2988	1.2131	1.1439	1.0898	1.0503	1.0243
0.55	1.6373	1.5016	1.3891	1.2958	1.2185	1.1549	1.1036	1.0640	1.0354	1.0170
0.5	1.4537	1.3607	1.2815	1.2146	1.1583	1.1118	1.0743	1.0456	1.0252	1.0122

Table S21. ARL_I values of the EWMA TBEA schemes for different statistics Z_i when the means of both A and T experience a shift based on $\lambda = 0.5$ when $\sigma_{0,A} = 1$ and $\sigma_{0,T} = 5$ with both variables A and T following the gamma distribution.

Z_1										
$\delta_T \backslash \delta_A$	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
0.95	32.5928	14.1247	7.8365	5.1081	3.7028	2.8854	2.3667	2.0163	1.7687	1.5878
0.9	21.6531	10.4582	6.2584	4.2974	3.2309	2.5853	2.1635	1.8725	1.6637	1.5096
0.85	15.0822	8.0248	5.1272	3.6814	2.8565	2.3395	1.9933	1.7501	1.5734	1.4419
0.8	10.9754	6.3547	4.2938	3.2031	2.5544	2.1357	1.8494	1.6454	1.4956	1.3833
0.75	8.3081	5.1716	3.6640	2.8242	2.3070	1.9649	1.7270	1.5554	1.4284	1.3326
0.7	6.5112	4.3081	3.1765	2.5186	2.1017	1.8206	1.6223	1.4779	1.3702	1.2886
0.65	5.2585	3.6601	2.7909	2.2682	1.9295	1.6976	1.5323	1.4109	1.3199	1.2505
0.6	4.3566	3.1608	2.4797	2.0599	1.7837	1.5923	1.4547	1.353	1.2763	1.2175
0.55	3.6874	2.7665	2.2238	1.8846	1.6592	1.5016	1.3876	1.3028	1.2385	1.1890
0.5	3.1768	2.4478	2.0100	1.7355	1.5522	1.4233	1.3294	1.2592	1.2056	1.1642
Z_2										
$\delta_T \backslash \delta_A$	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
0.95	78.7006	58.2493	44.1957	34.3040	27.1922	21.9786	18.0867	15.1314	12.8506	11.0635
0.9	46.5289	35.5624	27.8365	22.2663	18.1660	15.0898	12.7406	10.9169	9.4790	8.3290
0.85	28.8065	22.7281	18.3389	15.0978	12.6556	10.7812	9.3179	8.1576	7.2242	6.4632
0.8	18.6672	15.1873	12.6124	10.6656	9.1649	7.9875	7.0490	6.2898	5.6675	5.1512
0.75	12.6418	10.5862	9.0284	7.8234	6.8739	6.1135	5.4953	4.9860	4.5613	4.2032
0.7	8.9238	7.6730	6.7031	5.9362	5.3195	4.8160	4.3992	4.0500	3.7543	3.5014
0.65	6.5441	5.7620	5.1422	4.6421	4.2322	3.8916	3.6050	3.3613	3.1520	2.9706
0.6	4.9670	4.4661	4.0611	3.7281	3.4504	3.2160	3.0158	2.8433	2.6932	2.5617
0.55	3.8874	3.5602	3.2907	3.0653	2.8744	2.7110	2.5695	2.4461	2.3376	2.2415
0.5	3.1262	2.9093	2.7276	2.5734	2.4409	2.326	2.2254	2.1366	2.0578	1.9874
Z_3										
$\delta_T \backslash \delta_A$	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
0.95	99.8184	91.6553	83.5898	75.6929	68.0743	60.8640	54.1718	48.0510	42.4997	37.4891
0.9	57.1116	52.8137	48.5718	44.4250	40.4264	36.6325	33.0872	29.8106	26.8016	24.0492
0.85	34.2941	31.9655	29.6691	27.4252	25.2588	23.1944	21.2500	19.4336	17.7453	16.1811
0.8	21.6121	20.3159	19.0377	17.7877	16.5772	15.4172	14.3152	13.2748	12.2965	11.3791
0.75	14.2783	13.5380	12.8074	12.0912	11.3948	10.7230	10.0791	9.4650	8.8811	8.3274
0.7	9.8660	9.4329	9.0046	8.5833	8.1716	7.7716	7.3849	7.0125	6.6548	6.3120
0.65	7.1061	6.8469	6.5899	6.3362	6.0867	5.8427	5.6048	5.3736	5.1495	4.9326
0.6	5.3138	5.1556	4.9982	4.8421	4.6877	4.5357	4.3863	4.2400	4.0970	3.9574
0.55	4.1082	4.0100	3.9119	3.8141	3.7170	3.6207	3.5255	3.4315	3.3389	3.2478
0.5	3.2706	3.2088	3.1468	3.0849	3.0230	2.9613	2.8999	2.8389	2.7784	2.7185

Table S22. ARL_I values of the EWMA TBEA schemes for different statistics Z_i when the means of both A and T experience a shift based on $\lambda = 0.8$ when $\sigma_{0,A} = 1$ and $\sigma_{0,T} = 1$ with both variables A and T following the gamma distribution.

Z_1										
$\delta_T \backslash \delta_A$	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
0.95	13.1117	4.1029	2.0348	1.3657	1.1134	1.0268	1.0044	1.0005	1	1
0.9	6.8629	2.7558	1.6181	1.2091	1.0575	1.0115	1.0016	1.0002	1	1
0.85	4.1032	2.0339	1.3653	1.1135	1.0269	1.0045	1.0005	1	1	1
0.8	2.7548	1.6173	1.2089	1.0576	1.0117	1.0017	1.0002	1	1	1
0.75	2.0326	1.3647	1.1135	1.0272	1.0047	1.0006	1	1	1	1
0.7	1.6161	1.2087	1.0579	1.0119	1.0017	1.0002	1	1	1	1
0.65	1.3639	1.1136	1.0275	1.0048	1.0006	1.0001	1	1	1	1
0.6	1.2083	1.0582	1.0122	1.0019	1.0002	1	1	1	1	1
0.55	1.1137	1.0279	1.0051	1.0007	1.0001	1	1	1	1	1
0.5	1.0586	1.0127	1.0020	1.0003	1	1	1	1	1	1
Z_2										
$\delta_T \backslash \delta_A$	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
0.95	13.7689	4.9361	2.5366	1.6710	1.2931	1.1168	1.0398	1.0111	1.0025	1.0005
0.9	6.4819	3.0035	1.8494	1.3719	1.1526	1.0545	1.0162	1.0039	1.0008	1.0001
0.85	3.6597	2.0832	1.4710	1.1974	1.0736	1.0232	1.0060	1.0013	1.0002	1
0.8	2.3959	1.5971	1.2536	1.0981	1.0325	1.0090	1.0020	1.0004	1.0001	1
0.75	1.7597	1.3241	1.1292	1.0450	1.0132	1.0032	1.0006	1.0001	1	1
0.7	1.4131	1.1684	1.0612	1.0190	1.0049	1.0011	1.0002	1	1	1
0.65	1.2178	1.0822	1.0269	1.0074	1.0017	1.0003	1.0001	1	1	1
0.6	1.1090	1.0374	1.0110	1.0027	1.0006	1.0001	1	1	1	1
0.55	1.0512	1.0159	1.0042	1.0010	1.0002	1	1	1	1	1
0.5	1.0227	1.0064	1.0016	1.0003	1.0001	1	1	1	1	1
Z_3										
$\delta_T \backslash \delta_A$	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
0.95	13.6128	4.7768	2.3828	1.5407	1.1897	1.0490	1.0078	1.0007	1	1
0.9	6.3798	2.9827	1.8069	1.3183	1.1037	1.0235	1.0032	1.0002	1	1
0.85	3.5770	2.0893	1.4645	1.1769	1.0520	1.0102	1.0012	1.0001	1	1
0.8	2.3303	1.6045	1.2569	1.0912	1.0238	1.0041	1.0004	1	1	1
0.75	1.7084	1.3278	1.1334	1.0431	1.0100	1.0015	1.0001	1	1	1
0.7	1.3741	1.1689	1.0640	1.0187	1.0038	1.0005	1	1	1	1
0.65	1.1898	1.0812	1.0283	1.0074	1.0014	1.0002	1	1	1	1
0.6	1.0905	1.0362	1.0116	1.0028	1.0005	1.0001	1	1	1	1
0.55	1.0404	1.0151	1.0045	1.0010	1.0002	1	1	1	1	1
0.5	1.0170	1.0060	1.0017	1.0004	1.0001	1	1	1	1	1

Table S23. ARL_I values of the EWMA TBEA schemes for different statistics Z_i when the means of both A and T experience a shift based on $\lambda = 0.8$ when $\sigma_{0,A} = 1$ and $\sigma_{0,T} = 2$ with both variables A and T following the gamma distribution.

Z_1										
$\delta_T \backslash \delta_A$	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
0.95	22.9752	8.1981	3.9656	2.4119	1.7216	1.3747	1.1899	1.0914	1.0411	1.0172
0.9	13.2449	5.5145	3.0142	2.0006	1.5186	1.2675	1.1329	1.0623	1.0272	1.0110
0.85	8.2437	3.9556	2.4018	1.7159	1.3722	1.1894	1.0920	1.0420	1.0179	1.0071
0.8	5.5175	3.0014	1.9919	1.5142	1.2660	1.1330	1.0631	1.0281	1.0117	1.0046
0.75	3.9438	2.3893	1.7087	1.3690	1.1888	1.0926	1.0430	1.0187	1.0077	1.0030
0.7	2.9855	1.9807	1.5084	1.2640	1.1332	1.0641	1.0292	1.0125	1.0051	1.0020
0.65	2.3733	1.6992	1.3648	1.1879	1.0934	1.0442	1.0199	1.0085	1.0034	1.0013
0.6	1.9661	1.5008	1.2612	1.1332	1.0653	1.0306	1.0137	1.0058	1.0024	1.0009
0.55	1.6864	1.3590	1.1865	1.0942	1.0458	1.0214	1.0096	1.0041	1.0017	1.0007
0.5	1.4900	1.2571	1.1330	1.0669	1.0325	1.0152	1.0069	1.0030	1.0013	1.0005
Z_2										
$\delta_T \backslash \delta_A$	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
0.95	32.9736	15.4797	8.3864	5.1594	3.5282	2.6244	2.0839	1.7403	1.5117	1.3545
0.9	16.9478	8.9793	5.4222	3.6547	2.6898	2.1197	1.7610	1.5243	1.3626	1.2499
0.85	9.6390	5.7088	3.7901	2.7585	2.1569	1.7823	1.5371	1.3708	1.2554	1.1745
0.8	6.0214	3.9348	2.8307	2.1952	1.8039	1.5500	1.3790	1.2610	1.1785	1.1207
0.75	4.0893	2.9061	2.2346	1.8258	1.5629	1.3872	1.2665	1.1825	1.1238	1.0828
0.7	2.9847	2.2748	1.8477	1.5757	1.3952	1.272	1.1865	1.1269	1.0853	1.0566
0.65	2.3153	1.8693	1.5882	1.4030	1.2773	1.1905	1.1300	1.0878	1.0586	1.0386
0.6	1.8903	1.6000	1.4103	1.2824	1.1943	1.1331	1.0904	1.0608	1.0404	1.0265
0.55	1.6108	1.4170	1.2871	1.1980	1.1362	1.0931	1.0631	1.0423	1.0281	1.0185
0.5	1.4226	1.2911	1.2014	1.1392	1.0958	1.0655	1.0444	1.0299	1.0199	1.0131
Z_3										
$\delta_T \backslash \delta_A$	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
0.95	40.4935	21.2561	11.5367	6.6429	4.1305	2.7918	2.0432	1.6015	1.3288	1.1593
0.9	19.6529	11.5430	7.0395	4.5312	3.1071	2.2734	1.7671	1.4481	1.2430	1.1147
0.85	10.6948	6.9548	4.6899	3.3102	2.4560	1.9147	1.5631	1.3302	1.1767	1.0811
0.8	6.4663	4.5929	3.3671	2.5604	2.0230	1.6597	1.4109	1.2403	1.1265	1.0566
0.75	4.2900	3.2798	2.5729	2.0764	1.7252	1.4753	1.2974	1.1726	1.0893	1.0390
0.7	3.0805	2.5000	2.0702	1.7518	1.5156	1.3409	1.2131	1.1226	1.0623	1.0267
0.65	2.3631	2.0116	1.7390	1.5283	1.3663	1.2431	1.1515	1.0862	1.0433	1.0183
0.6	1.9148	1.6929	1.5144	1.3720	1.2595	1.1723	1.107	1.0604	1.0301	1.0126
0.55	1.6235	1.4790	1.3596	1.2619	1.1834	1.1218	1.0754	1.0424	1.0210	1.0088
0.5	1.4292	1.3331	1.2521	1.1846	1.1297	1.0862	1.0534	1.0301	1.0150	1.0064

Table S24. ARL_I values of the EWMA TBEA schemes for different statistics Z_i when the means of both A and T experience a shift based on $\lambda = 0.8$ when $\sigma_{0,A} = 1$ and $\sigma_{0,T} = 5$ with both variables A and T following the gamma distribution.

Z_1										
$\delta_T \backslash \delta_A$	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
0.95	35.5282	14.9018	7.9788	5.0439	3.5723	2.7403	2.2275	1.8908	1.6593	1.4946
0.9	24.1151	11.0119	6.3040	4.1928	3.0862	2.4386	2.0288	1.7542	1.5625	1.4245
0.85	16.8712	8.3580	5.0914	3.5473	2.7041	2.1951	1.8652	1.6402	1.4809	1.3650
0.8	12.1625	6.5067	4.1967	3.0500	2.4002	1.9967	1.7296	1.5444	1.4117	1.3143
0.75	9.0272	5.1871	3.5243	2.6614	2.1557	1.8337	1.6165	1.4638	1.3531	1.2711
0.7	6.8886	4.2265	3.0104	2.3536	1.9570	1.6989	1.5218	1.3956	1.3033	1.2343
0.65	5.3948	3.5129	2.6113	2.1068	1.7942	1.5866	1.4421	1.3379	1.2609	1.2029
0.6	4.3269	2.9727	2.2968	1.9070	1.6597	1.4927	1.3748	1.2889	1.2248	1.1761
0.55	3.5458	2.5564	2.0457	1.7435	1.5480	1.4137	1.3178	1.2471	1.1939	1.1532
0.5	2.9620	2.2301	1.8427	1.6086	1.4544	1.3470	1.2693	1.2114	1.1675	1.1336
Z_2										
$\delta_T \backslash \delta_A$	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
0.95	82.9631	63.2469	49.3810	39.3603	31.9465	26.348	22.0435	18.6804	16.0150	13.8748
0.9	49.9000	39.1255	31.3542	25.6054	21.2592	17.9103	15.2864	13.1996	11.5177	10.1459
0.85	31.2312	25.1547	20.6656	17.2707	14.6509	12.5933	10.9521	9.6247	8.5380	7.6382
0.8	20.3220	16.7913	14.1233	12.0630	10.4421	9.1459	8.0943	7.2304	6.5125	5.9097
0.75	13.7323	11.6229	9.9945	8.7121	7.6846	6.8489	6.1602	5.5859	5.1020	4.6905
0.7	9.6230	8.3298	7.3114	6.4944	5.8286	5.2784	4.8183	4.4293	4.0973	3.8115
0.65	6.9811	6.1696	5.5185	4.9872	4.5473	4.1784	3.8656	3.5978	3.3665	3.1652
0.6	5.2326	4.7130	4.2888	3.9370	3.6414	3.3902	3.1745	2.9877	2.8245	2.6810
0.55	4.0435	3.7051	3.4243	3.1881	2.9869	2.8137	2.6633	2.5316	2.4155	2.3123
0.5	3.2140	2.9907	2.8027	2.6424	2.5042	2.3838	2.2782	2.1848	2.1017	2.0273
Z_3										
$\delta_T \backslash \delta_A$	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
0.95	104.9545	98.7788	92.7768	86.9474	81.2958	75.8351	70.5858	65.5708	60.8076	56.3023
0.9	61.3329	58.0682	54.8802	51.7700	48.7416	45.8025	42.9632	40.2347	37.6254	35.1386
0.85	37.3846	35.6002	33.8504	32.1362	30.4602	28.8267	27.2408	25.7082	24.2328	22.8166
0.8	23.7450	22.7390	21.7486	20.7747	19.8188	18.8830	17.9703	17.0833	16.2241	15.3938
0.75	15.6969	15.1132	14.5365	13.9673	13.4066	12.8555	12.3154	11.7877	11.2735	10.7735
0.7	10.7837	10.4361	10.0915	9.7503	9.4128	9.0798	8.7520	8.4301	8.1147	7.8062
0.65	7.6854	7.4735	7.2627	7.0533	6.8456	6.6398	6.4363	6.2355	6.0377	5.8432
0.6	5.6703	5.5384	5.4069	5.2758	5.1453	5.0156	4.8868	4.7591	4.6328	4.5079
0.55	4.3211	4.2376	4.1542	4.0707	3.9874	3.9043	3.8214	3.7390	3.6570	3.5756
0.5	3.3928	3.3393	3.2857	3.2319	3.178	3.1241	3.0702	3.0163	2.9625	2.9089

Table S25. ARL_I values of the EWMA TBEA schemes for different statistics Z_i when the means of both A and T experience a shift based on $\lambda = 1$ when $\sigma_{0,A} = 1$ and $\sigma_{0,T} = 1$ with both variables A and T following the gamma distribution.

Z_1										
$\delta_T \backslash \delta_A$	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
0.95	15.8734	4.8574	2.1911	1.3787	1.1071	1.0237	1.0038	1.0004	1	1
0.9	8.3531	3.1155	1.6749	1.2061	1.0524	1.0100	1.0013	1.0001	1	1
0.85	4.8593	2.1899	1.3781	1.1072	1.0239	1.0039	1.0004	1	1	1
0.8	3.1148	1.6739	1.2059	1.0526	1.0101	1.00144	1.0001	1	1	1
0.75	2.1883	1.3774	1.1072	1.0241	1.0040	1.0005	1	1	1	1
0.7	1.6724	1.2056	1.0528	1.0103	1.0015	1.0002	1	1	1	1
0.65	1.3764	1.1073	1.0244	1.0041	1.0005	1	1	1	1	1
0.6	1.2051	1.0532	1.0106	1.0016	1.0002	1	1	1	1	1
0.55	1.1074	1.0249	1.0044	1.0006	1.0001	1	1	1	1	1
0.5	1.0537	1.0111	1.0017	1.0002	1	1	1	1	1	1
Z_2										
$\delta_T \backslash \delta_A$	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
0.95	16.7306	5.997	2.8543	1.7389	1.2917	1.1061	1.0335	1.0088	1.0019	1.0003
0.9	7.7587	3.4180	1.9518	1.3797	1.1424	1.0472	1.0132	1.003	1.0006	1.0001
0.85	4.1786	2.2273	1.4916	1.1891	1.0654	1.0194	1.0048	1.0010	1.0002	1
0.8	2.5875	1.6341	1.2487	1.0894	1.0279	1.0073	1.0016	1.0003	1	1
0.75	1.8166	1.3244	1.1205	1.0395	1.0110	1.0026	1.0005	1.0001	1	1
0.7	1.4205	1.1605	1.0550	1.0162	1.0040	1.0008	1.0001	1	1	1
0.65	1.2115	1.0754	1.0235	1.0062	1.0014	1.0003	1	1	1	1
0.6	1.1020	1.0334	1.0094	1.0023	1.0005	1.0001	1	1	1	1
0.55	1.0466	1.0139	1.0036	1.0008	1.0002	1	1	1	1	1
0.5	1.0203	1.0056	1.0013	1.0003	1.0001	1	1	1	1	1
Z_3										
$\delta_T \backslash \delta_A$	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
0.95	16.9576	5.9424	2.6871	1.5787	1.1752	1.0397	1.0057	1.0004	1	1
0.9	7.7666	3.4393	1.9056	1.3149	1.0903	1.0183	1.0023	1.0002	1	1
0.85	4.1116	2.2448	1.4815	1.1637	1.0433	1.0078	1.0008	1	1	1
0.8	2.5122	1.6410	1.2483	1.0800	1.0191	1.0030	1.0003	1	1	1
0.75	1.7539	1.3248	1.1218	1.0363	1.0078	1.0011	1.0001	1	1	1
0.7	1.3743	1.1583	1.0560	1.0153	1.0029	1.0004	1	1	1	1
0.65	1.1803	1.0730	1.0240	1.0060	1.0010	1.0001	1	1	1	1
0.6	1.0827	1.0315	1.0096	1.0022	1.0004	1	1	1	1	1
0.55	1.0359	1.0129	1.0037	1.0008	1.0001	1	1	1	1	1
0.5	1.0148	1.0050	1.0014	1.0003	1	1	1	1	1	1

Table S26. ARL_I values of the EWMA TBEA schemes for different statistics Z_i when the means of both A and T experience a shift based on $\lambda = 1$ when $\sigma_{0,A} = 1$ and $\sigma_{0,T} = 2$ with both variables A and T following the gamma distribution.

Z_1										
$\delta_T \backslash \delta_A$	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
0.95	26.7089	9.7085	4.5228	2.5998	1.7728	1.3801	1.1841	1.0855	1.0375	1.0153
0.9	15.7244	6.4494	3.3428	2.1020	1.5397	1.2648	1.1264	1.0575	1.0246	1.0098
0.85	9.8111	4.5186	2.5880	1.7656	1.3772	1.1836	1.0861	1.0383	1.0160	1.0063
0.8	6.4772	3.3301	2.0911	1.5343	1.2631	1.1266	1.0583	1.0254	1.0104	1.0040
0.75	4.5151	2.5732	1.7566	1.3734	1.1829	1.0869	1.0394	1.0169	1.0068	1.0026
0.7	3.3146	2.0773	1.5273	1.2609	1.1268	1.0594	1.0265	1.0113	1.0045	1.0017
0.65	2.5546	1.7448	1.3685	1.1820	1.0877	1.0406	1.0180	1.0076	1.0031	1.0012
0.6	2.0593	1.5181	1.2578	1.1269	1.0607	1.0280	1.0123	1.0052	1.0021	1.0008
0.55	1.7291	1.3617	1.1805	1.0887	1.0423	1.0195	1.0086	1.0037	1.0015	1.0006
0.5	1.5052	1.2533	1.1268	1.0623	1.0298	1.0138	1.0062	1.0027	1.0011	1.0005
Z_2										
$\delta_T \backslash \delta_A$	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
0.95	36.8179	18.1399	10.0276	6.1289	4.0873	2.9391	2.2545	1.8271	1.5508	1.3678
0.9	19.2283	10.4581	6.3179	4.1787	2.9874	2.2823	1.8444	1.5622	1.3758	1.2505
0.85	10.9358	6.5211	4.2742	3.0368	2.3102	1.8615	1.5736	1.3837	1.2563	1.1701
0.8	6.7412	4.3744	3.0870	2.3380	1.8783	1.5846	1.3915	1.2621	1.1745	1.1151
0.75	4.4800	3.1383	2.3655	1.8946	1.5953	1.3990	1.2677	1.1788	1.1185	1.0776
0.7	3.1904	2.3925	1.9103	1.6054	1.4062	1.2731	1.1831	1.1219	1.0804	1.0523
0.65	2.4188	1.9251	1.6148	1.4129	1.2783	1.1873	1.1253	1.0832	1.0546	1.0354
0.6	1.9386	1.6233	1.4190	1.2831	1.1913	1.1287	1.0860	1.0570	1.0373	1.0242
0.55	1.6304	1.4242	1.2874	1.1951	1.1321	1.0889	1.0594	1.0394	1.0258	1.0168
0.5	1.4282	1.2910	1.1985	1.1353	1.0919	1.0621	1.0416	1.0277	1.0182	1.0119
Z_3										
$\delta_T \backslash \delta_A$	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
0.95	45.8496	26.0913	15.0777	8.9173	5.4498	3.4816	2.3552	1.7088	1.3429	1.1456
0.9	22.6539	14.0590	8.8646	5.7180	3.8050	2.6377	1.9249	1.4935	1.2400	1.1009
0.85	12.3220	8.2695	5.6388	3.9302	2.8199	2.0996	1.6359	1.3433	1.1668	1.0693
0.8	7.3343	5.2788	3.8588	2.8794	2.2063	1.7471	1.4388	1.2382	1.1152	1.0472
0.75	4.7456	3.6332	2.8226	2.2348	1.8117	1.5112	1.3030	1.1646	1.0791	1.0319
0.7	3.3135	2.6768	2.1918	1.8253	1.5516	1.3512	1.2091	1.1134	1.0541	1.0216
0.65	2.4770	2.0952	1.7936	1.5581	1.3772	1.2418	1.1442	1.0780	1.0370	1.0147
0.6	1.9663	1.7285	1.5353	1.3805	1.2591	1.1668	1.0996	1.0538	1.0254	1.0101
0.55	1.6434	1.4910	1.3644	1.2611	1.1788	1.1154	1.0690	1.0373	1.0177	1.0071
0.5	1.4341	1.3344	1.2501	1.1803	1.1241	1.0805	1.0484	1.0263	1.0126	1.0051

Table S27. ARL_I values of the EWMA TBEA schemes for different statistics Z_i when the means of both A and T experience a shift based on $\lambda = 1$ when $\sigma_{0,A} = 1$ and $\sigma_{0,T} = 5$ with both variables A and T following the gamma distribution.

Z_1										
$\delta_T \backslash \delta_A$	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
0.95	39.3985	16.5533	8.7093	5.3809	3.7293	2.8105	2.2545	1.8962	1.6543	1.4851
0.9	27.3847	12.2917	6.8472	4.4349	3.1938	2.4827	2.0421	1.7527	1.5543	1.4138
0.85	19.4449	9.3147	5.4839	3.7156	2.7743	2.2199	1.8687	1.6339	1.4707	1.3538
0.8	14.1018	7.2008	4.4718	3.1622	2.4425	2.0076	1.7263	1.5353	1.4007	1.3032
0.75	10.4424	5.6761	3.7102	2.7319	2.1779	1.8349	1.6089	1.4530	1.3418	1.2605
0.7	7.8929	4.5598	3.1301	2.3938	1.9651	1.6937	1.5116	1.3842	1.2923	1.2244
0.65	6.0872	3.7308	2.6829	2.1258	1.7928	1.5775	1.4306	1.3265	1.2505	1.1938
0.6	4.7879	3.1070	2.3346	1.9115	1.6523	1.4815	1.363	1.2779	1.2151	1.1679
0.55	3.8389	2.6317	2.0604	1.7388	1.5371	1.4017	1.3064	1.2370	1.1852	1.1459
0.5	3.1356	2.2651	1.8427	1.5985	1.4421	1.3351	1.2586	1.2023	1.1598	1.1271
Z_2										
$\delta_T \backslash \delta_A$	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
0.95	84.4896	65.0565	51.3038	41.2951	33.8334	28.1527	23.7477	20.2759	17.4998	15.2510
0.9	51.1717	40.4912	32.7386	26.9637	22.5651	19.1494	16.4515	14.2884	12.5308	11.0857
0.85	32.1958	26.1402	21.6385	18.2110	15.5472	13.4397	11.7462	10.3664	9.2284	8.2795
0.8	21.0184	17.4832	14.7955	12.7068	11.0526	9.7209	8.6333	7.7339	6.9816	6.3462
0.75	14.2195	12.0989	10.4524	9.1482	8.0969	7.2367	6.5237	5.9257	5.4191	4.9858
0.7	9.9557	8.6515	7.6190	6.7864	6.1042	5.5376	5.0613	4.6567	4.3098	4.0098
0.65	7.2033	6.3831	5.7220	5.1799	4.7291	4.3494	4.0261	3.7481	3.5072	3.2966
0.6	5.3777	4.8518	4.4207	4.0619	3.7593	3.5012	3.2787	3.0854	2.9161	2.7668
0.55	4.1357	3.7931	3.5079	3.2672	3.0615	2.8840	2.7295	2.5937	2.4738	2.3670
0.5	3.2706	3.0447	2.8540	2.6909	2.5500	2.4271	2.3189	2.2231	2.1377	2.0611
Z_3										
$\delta_T \backslash \delta_A$	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
0.95	106.4812	100.9187	95.5709	90.4327	85.4988	80.7642	76.2237	71.8724	67.7053	63.7175
0.9	62.6957	59.7610	56.9235	54.1815	51.5331	48.9764	46.5098	44.1314	41.8394	39.6320
0.85	38.4433	36.8410	35.2836	33.7704	32.3009	30.8743	29.4901	28.1477	26.8465	25.5858
0.8	24.5158	23.6124	22.7299	21.8681	21.0268	20.2059	19.4051	18.6243	17.8633	17.1219
0.75	16.2371	15.7122	15.197	14.6916	14.1958	13.7097	13.2331	12.7661	12.3086	11.8606
0.7	11.1523	10.8387	10.5297	10.2252	9.9251	9.6296	9.3385	9.0520	8.7700	8.4925
0.65	7.9310	7.7390	7.5490	7.3610	7.1750	6.9911	6.8091	6.6293	6.4515	6.2757
0.6	5.8301	5.7100	5.5906	5.4721	5.3544	5.2375	5.1215	5.0063	4.8920	4.7785
0.55	4.4223	4.3457	4.2694	4.1933	4.1175	4.042	3.9667	3.8917	3.8171	3.7427
0.5	3.4548	3.4053	3.3558	3.3063	3.2568	3.2074	3.1580	3.1086	3.0592	3.0099

Table S28. ARL_I values of the EWMA TBEA schemes for different statistics Z_i when the means of both A and T experience a shift based on $\lambda = 0.1$ when $\sigma_{0,A} = 1$ and $\sigma_{0,T} = 1$ with variables A following the normal distribution and T following the gamma distribution.

Z_1										
$\delta_T \backslash \delta_A$	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
0.95	6.6646	3.3049	2.2571	1.7524	1.4138	1.1757	1.0528	1.0110	1.0016	1.0002
0.9	4.4078	2.6669	1.9707	1.5708	1.2805	1.1009	1.0253	1.0044	1.0005	1
0.85	3.3041	2.2571	1.7522	1.4134	1.1756	1.0531	1.0112	1.0016	1.0002	1
0.8	2.6665	1.9707	1.5704	1.2801	1.1010	1.0256	1.0046	1.0006	1.0001	1
0.75	2.2569	1.7519	1.4128	1.1755	1.0533	1.0114	1.0017	1.0002	1	1
0.7	1.9706	1.5698	1.2796	1.1012	1.0259	1.0047	1.0006	1.0001	1	1
0.65	1.7516	1.4120	1.1753	1.0537	1.0117	1.0018	1.0002	1	1	1
0.6	1.5690	1.2790	1.1014	1.0264	1.0049	1.0007	1.0001	1	1	1
0.55	1.4110	1.1751	1.0542	1.0121	1.0020	1.0002	1	1	1	1
0.5	1.2780	1.1016	1.0270	1.0053	1.0008	1.0001	1	1	1	1
Z_2										
$\delta_T \backslash \delta_A$	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
0.95	7.3970	4.1693	2.9664	2.3533	1.9952	1.7558	1.5503	1.3587	1.2028	1.0985
0.9	4.9800	3.2712	2.4929	2.0610	1.787	1.5702	1.3747	1.2146	1.1058	1.0448
0.85	3.7025	2.6749	2.1449	1.8258	1.5921	1.3912	1.2269	1.1136	1.0490	1.0182
0.8	2.9195	2.2529	1.8746	1.6172	1.4084	1.2395	1.1217	1.0534	1.0203	1.0067
0.75	2.3940	1.9368	1.6468	1.4267	1.2525	1.1301	1.0581	1.0225	1.0076	1.0023
0.7	2.0168	1.6828	1.4467	1.2660	1.1390	1.0631	1.0250	1.0087	1.0026	1.0007
0.65	1.7277	1.4692	1.2802	1.1482	1.0685	1.0276	1.0098	1.0031	1.0009	1.0002
0.6	1.4955	1.2953	1.1578	1.0741	1.0306	1.0111	1.0036	1.0011	1.0003	1.0001
0.55	1.3118	1.1679	1.0801	1.0338	1.0126	1.0042	1.0013	1.0004	1.0001	1
0.5	1.1785	1.0865	1.0373	1.0143	1.005	1.0016	1.0005	1.0001	1	1
Z_3										
$\delta_T \backslash \delta_A$	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
0.95	7.4624	4.2179	2.9967	2.3702	2.0270	1.8232	1.5974	1.3354	1.1313	1.0330
0.9	5.0911	3.3886	2.5881	2.1443	1.8925	1.6868	1.4444	1.2159	1.0724	1.0156
0.85	3.8063	2.8099	2.2721	1.9599	1.7469	1.5272	1.2990	1.1253	1.0361	1.0067
0.8	3.0081	2.3882	2.0237	1.7873	1.5800	1.3652	1.1805	1.0654	1.0164	1.0027
0.75	2.4695	2.0702	1.8110	1.6059	1.4069	1.2260	1.0975	1.0308	1.0068	1.0010
0.7	2.0834	1.8129	1.6077	1.4226	1.2541	1.1245	1.0473	1.0132	1.0026	1.0003
0.65	1.7858	1.5857	1.4143	1.2622	1.1409	1.0614	1.0208	1.0052	1.0009	1.0001
0.6	1.5398	1.3849	1.2516	1.1443	1.0699	1.0275	1.0084	1.0019	1.0003	1
0.55	1.3385	1.2260	1.1358	1.0712	1.0315	1.0114	1.0032	1.0007	1.0001	1
0.5	1.1906	1.1187	1.0661	1.0321	1.0132	1.0045	1.0012	1.0002	1	1

Table S29. ARL_I values of the EWMA TBEA schemes for different statistics Z_i when the means of both A and T experience a shift based on $\lambda = 0.1$ when $\sigma_{0,A} = 1$ and $\sigma_{0,T} = 2$ with variables A following the normal distribution and T following the gamma distribution.

Z_1										
$\delta_T \backslash \delta_A$	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
0.95	12.5307	5.7031	3.6262	2.6911	2.1690	1.8269	1.5713	1.3701	1.2192	1.1174
0.9	7.9425	4.4250	3.0804	2.3976	1.9830	1.6907	1.4635	1.2876	1.1623	1.0830
0.85	5.6915	3.6214	2.6895	2.1686	1.8261	1.5695	1.3683	1.2183	1.1176	1.0577
0.8	4.4165	3.0771	2.3967	1.9825	1.6891	1.4614	1.2861	1.1620	1.0836	1.0395
0.75	3.6153	2.6875	2.1681	1.8249	1.5673	1.3660	1.2172	1.1178	1.0585	1.0268
0.7	3.0729	2.3956	1.9818	1.6871	1.4586	1.2840	1.1615	1.0842	1.0406	1.0182
0.65	2.6849	2.1675	1.8235	1.5643	1.3631	1.2157	1.1179	1.0596	1.0280	1.0124
0.6	2.3943	1.9810	1.6845	1.4548	1.2813	1.1607	1.0850	1.0419	1.0194	1.0085
0.55	2.1669	1.8216	1.5602	1.3590	1.2136	1.1180	1.0609	1.0296	1.0136	1.0060
0.5	1.9801	1.6807	1.4495	1.2774	1.1595	1.0859	1.0436	1.0211	1.0097	1.0043
Z_2										
$\delta_T \backslash \delta_A$	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
0.95	14.8736	7.9893	5.4283	4.1306	3.3585	2.8516	2.4959	2.2341	2.0341	1.8763
0.9	9.3717	5.9436	4.3645	3.4737	2.9085	2.5214	2.2412	2.0299	1.8647	1.7305
0.85	6.6215	4.6504	3.6083	2.9731	2.5495	2.2490	2.0254	1.8524	1.7134	1.5976
0.8	5.0061	3.7672	3.0466	2.5807	2.2572	2.0205	1.8396	1.6959	1.5777	1.4774
0.75	3.9566	3.1306	2.6153	2.2662	2.0152	1.8262	1.678	1.5576	1.457	1.3714
0.7	3.2272	2.6536	2.2756	2.0094	1.8121	1.6596	1.5375	1.4369	1.3526	1.2815
0.65	2.6959	2.2855	2.0030	1.7973	1.6408	1.5173	1.4171	1.3344	1.2656	1.2085
0.6	2.2955	1.9957	1.7816	1.6214	1.4969	1.3975	1.3166	1.2503	1.1959	1.1516
0.55	1.9870	1.7647	1.6013	1.4764	1.3781	1.2994	1.2356	1.184	1.1422	1.1089
0.5	1.7462	1.5803	1.4555	1.359	1.2827	1.2217	1.1728	1.1336	1.1025	1.0779
Z_3										
$\delta_T \backslash \delta_A$	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
0.95	15.1162	8.0469	5.4305	4.1182	3.3430	2.8365	2.4794	2.2141	2.0278	1.9118
0.9	9.4634	6.1095	4.5079	3.5915	3.0064	2.6038	2.3105	2.0934	1.9434	1.8396
0.85	6.6909	4.8508	3.8133	3.1588	2.7139	2.3943	2.1563	1.9808	1.8554	1.7550
0.8	5.0741	3.9708	3.2726	2.7979	2.4581	2.2054	2.0142	1.8708	1.7601	1.6596
0.75	4.0238	3.3234	2.8410	2.4930	2.2330	2.0340	1.8802	1.7596	1.6575	1.5581
0.7	3.2912	2.8292	2.4898	2.2328	2.0336	1.8767	1.7511	1.6465	1.5515	1.4568
0.65	2.7547	2.4417	2.1996	2.0087	1.8556	1.7308	1.6265	1.5347	1.4480	1.3621
0.6	2.3479	2.1318	1.9573	1.8146	1.6961	1.596	1.5087	1.4289	1.3528	1.2789
0.55	2.0324	1.8809	1.754	1.6467	1.5546	1.4741	1.4016	1.3342	1.2702	1.2096
0.5	1.7844	1.6771	1.5843	1.5033	1.4319	1.3677	1.3088	1.2539	1.2024	1.1548

Table S30. ARL_I values of the EWMA TBEA schemes for different statistics Z_i when the means of both A and T experience a shift based on $\lambda = 0.1$ when $\sigma_{0,A} = 1$ and $\sigma_{0,T} = 5$ with variables A following the normal distribution and T following the gamma distribution.

Z_1										
$\delta_T \backslash \delta_A$	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
0.95	30.3943	14.4216	8.7313	6.1087	6.1087	3.7993	3.2121	2.7891	2.4661	2.2086
0.9	20.2845	10.9661	7.1843	5.2806	5.2806	3.4723	2.9822	2.6168	2.3296	2.0964
0.85	14.5062	8.7028	6.0712	4.6459	4.6459	3.2020	2.7861	2.4650	2.2062	1.9929
0.8	10.9716	7.1459	5.2426	4.1481	4.1481	2.9757	2.6162	2.3291	2.0929	1.8967
0.75	8.6761	6.0291	4.6079	3.7504	3.7504	2.7835	2.4665	2.2053	1.9877	1.8068
0.7	7.1062	5.1987	4.1102	3.4281	3.4281	2.6180	2.3322	2.0908	1.8891	1.7223
0.65	5.9838	4.5627	3.7125	3.1639	3.1639	2.4729	2.2091	1.9834	1.7960	1.6430
0.6	5.1504	4.0633	3.3902	2.9456	2.9456	2.3430	2.094	1.8813	1.7078	1.5685
0.55	4.5114	3.6634	3.1264	2.7641	2.7641	2.2240	1.9844	1.7836	1.6242	1.4988
0.5	4.0080	3.3384	2.9089	2.6123	2.6123	2.1122	1.8784	1.6896	1.5450	1.4338
Z_2										
$\delta_T \backslash \delta_A$	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
0.95	69.5784	48.5209	35.2561	26.6283	20.8322	16.8091	13.9256	11.7958	10.1796	8.9234
0.9	41.0117	30.1632	23.0293	18.1863	14.7934	12.3423	10.5198	9.1288	8.0423	7.1761
0.85	25.7189	19.8421	15.8160	12.9711	10.9004	9.3505	8.1609	7.2269	6.479	5.8697
0.8	17.0408	13.7075	11.3345	9.5952	8.2852	7.2741	6.4763	5.8346	5.3096	4.8738
0.75	11.8439	9.8734	8.4204	7.3199	6.4659	5.7888	5.2418	4.7925	4.4181	4.1021
0.7	8.5763	7.3686	6.4494	5.7327	5.1620	4.6989	4.3171	3.9977	3.7272	3.4955
0.65	6.4311	5.6679	5.0704	4.5927	4.2037	3.8816	3.6113	3.3815	3.1841	3.0129
0.6	4.9690	4.4744	4.0777	3.7534	3.4841	3.2573	3.0639	2.8973	2.7523	2.625
0.55	3.9397	3.6131	3.3454	3.1224	2.934	2.7729	2.6336	2.5121	2.4052	2.3105
0.5	3.1951	2.9765	2.7940	2.6394	2.5068	2.3919	2.2914	2.2027	2.1239	2.0534
Z_3										
$\delta_T \backslash \delta_A$	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
0.95	70.6376	52.7301	39.0314	29.2811	22.5437	17.8696	14.5656	12.1736	10.3975	9.0457
0.9	37.8309	30.1106	24.0603	19.4854	16.0598	13.4785	11.5085	9.9819	8.78	7.8191
0.85	22.8039	19.2137	16.2824	13.9285	12.0440	10.5288	9.3013	8.2982	7.4708	6.7821
0.8	15.0980	13.2942	11.7561	10.4553	9.3570	8.4281	7.6397	6.9675	6.3919	5.8965
0.75	10.7063	9.7371	8.8781	8.1205	7.4537	6.8669	6.35	5.8941	5.4914	5.135
0.7	7.9737	7.4248	6.9224	6.4645	6.0480	5.6698	5.3266	5.0154	4.7333	4.4775
0.65	6.1533	5.8300	5.5265	5.2426	4.9776	4.7308	4.5014	4.2885	4.0912	3.9087
0.6	4.8764	4.6808	4.4935	4.3145	4.144	3.982	3.8283	3.6829	3.5457	3.4165
0.55	3.9457	3.8253	3.7082	3.5944	3.4841	3.3776	3.275	3.1762	3.0815	2.991
0.5	3.2477	3.1729	3.0992	3.0267	2.9555	2.8857	2.8176	2.7511	2.6865	2.6239

Table S31. ARL_I values of the EWMA TBEA schemes for different statistics Z_i when the means of both A and T experience a shift based on $\lambda = 0.2$ when $\sigma_{0,A} = 1$ and $\sigma_{0,T} = 1$ with variables A following the normal distribution and T following the gamma distribution.

Z_1										
$\delta_T \backslash \delta_A$	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
0.95	6.8973	3.1644	2.0861	1.5788	1.2693	1.0939	1.023	1.0039	1.0005	1
0.9	4.3499	2.5027	1.7965	1.406	1.1658	1.0487	1.0100	1.0014	1.0001	1
0.85	3.1632	2.0858	1.5784	1.2690	1.0940	1.0232	1.0040	1.0005	1	1
0.8	2.5021	1.7961	1.4056	1.1658	1.0490	1.0102	1.0015	1.0002	1	1
0.75	2.0853	1.5779	1.2687	1.0942	1.0235	1.0041	1.0005	1	1	1
0.7	1.7956	1.4050	1.1657	1.0493	1.0104	1.0016	1.0002	1	1	1
0.65	1.5772	1.2682	1.0944	1.0238	1.0043	1.0006	1.0001	1	1	1
0.6	1.4041	1.1655	1.0497	1.0107	1.0017	1.0002	1	1	1	1
0.55	1.2675	1.0947	1.0244	1.0046	1.0006	1.0001	1	1	1	1
0.5	1.1653	1.0503	1.0112	1.0019	1.0002	1	1	1	1	1
Z_2										
$\delta_T \backslash \delta_A$	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
0.95	7.1877	3.7231	2.5690	2.0135	1.6815	1.4381	1.2485	1.1193	1.048	1.0161
0.9	4.5365	2.8540	2.1389	1.7455	1.4791	1.2774	1.1373	1.0571	1.0199	1.0059
0.85	3.2634	2.3043	1.8244	1.5249	1.3085	1.1570	1.0675	1.0244	1.0075	1.0019
0.8	2.5293	1.9242	1.5780	1.3424	1.1786	1.0793	1.0297	1.0094	1.0026	1.0006
0.75	2.0540	1.6414	1.3800	1.2023	1.0925	1.0359	1.0118	1.0033	1.0008	1.0002
0.7	1.7197	1.4226	1.2284	1.1074	1.0431	1.0147	1.0043	1.0011	1.0002	1
0.65	1.4724	1.2572	1.1240	1.0514	1.0183	1.0056	1.0015	1.0003	1.0001	1
0.6	1.2897	1.1425	1.0609	1.0225	1.0072	1.002	1.0005	1.0001	1	1
0.55	1.1631	1.0718	1.0274	1.0091	1.0027	1.0007	1.0002	1	1	1
0.5	1.0840	1.0333	1.0116	1.0036	1.001	1.0002	1.0001	1	1	1
Z_3										
$\delta_T \backslash \delta_A$	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
0.95	7.1585	3.7154	2.5619	2.0177	1.6931	1.415	1.1825	1.0529	1.0094	1.0010
0.9	4.5830	2.9227	2.2015	1.8098	1.5270	1.2791	1.1055	1.0262	1.0040	1.0004
0.85	3.3157	2.3969	1.9212	1.6167	1.3673	1.1695	1.0552	1.0118	1.0016	1.0001
0.8	2.5764	2.0218	1.6862	1.4354	1.2308	1.0925	1.0261	1.0049	1.0006	1
0.75	2.0943	1.7334	1.4797	1.2782	1.1299	1.0454	1.0112	1.0019	1.0002	1
0.7	1.7511	1.4989	1.3055	1.1589	1.0655	1.0202	1.0044	1.0007	1.0001	1
0.65	1.4916	1.3113	1.1743	1.0812	1.0298	1.0082	1.0016	1.0002	1	1
0.6	1.2968	1.1749	1.0890	1.0374	1.0125	1.0031	1.0006	1.0001	1	1
0.55	1.1622	1.0883	1.0412	1.0159	1.0049	1.0011	1.0002	1	1	1
0.5	1.0804	1.0407	1.0177	1.0064	1.0018	1.0004	1.0001	1	1	1

Table S32. ARL_I values of the EWMA TBEA schemes for different statistics Z_i when the means of both A and T experience a shift based on $\lambda = 0.2$ when $\sigma_{0,A} = 1$ and $\sigma_{0,T} = 2$ with variables A following the normal distribution and T following the gamma distribution.

Z_1										
$\delta_T \backslash \delta_A$	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
0.95	13.3577	5.6744	3.4530	2.4960	1.9727	1.6371	1.4017	1.2359	1.1267	1.0618
0.9	8.1643	4.2930	2.8905	2.2003	1.7880	1.5086	1.3103	1.1747	1.0898	1.0421
0.85	5.6620	3.4472	2.4933	1.9708	1.6351	1.3999	1.2350	1.1268	1.0625	1.0283
0.8	4.2834	2.8862	2.1980	1.7858	1.5063	1.3087	1.1743	1.0903	1.0430	1.0189
0.75	3.4400	2.4900	1.9684	1.6325	1.3976	1.2338	1.1269	1.0634	1.0292	1.0125
0.7	2.8809	2.1951	1.7830	1.5033	1.3067	1.1737	1.0909	1.0440	1.0198	1.0084
0.65	2.4859	1.9653	1.6290	1.3945	1.2322	1.1269	1.0644	1.0305	1.0135	1.0057
0.6	2.1914	1.7792	1.4993	1.3038	1.1728	1.0916	1.0454	1.0211	1.0093	1.0039
0.55	1.9612	1.6242	1.3902	1.2300	1.1269	1.0656	1.0320	1.0148	1.0065	1.0028
0.5	1.7739	1.4935	1.2997	1.1715	1.0924	1.0471	1.0228	1.0106	1.0047	1.0020
Z_2										
$\delta_T \backslash \delta_A$	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
0.95	16.1265	7.7633	4.9705	3.6733	2.9424	2.4787	2.1604	1.9286	1.7515	1.6103
0.9	9.2794	5.5006	3.9038	3.0543	2.5348	2.1872	1.9389	1.7522	1.6056	1.4863
0.85	6.1977	4.1864	3.1854	2.5988	2.2173	1.9506	1.7535	1.6011	1.479	1.3788
0.8	4.5379	3.3405	2.6721	2.2512	1.9638	1.7554	1.5969	1.472	1.3709	1.2881
0.75	3.5254	2.7562	2.2893	1.9786	1.7579	1.5930	1.4651	1.3631	1.2808	1.2141
0.7	2.8530	2.3319	1.9950	1.7609	1.5893	1.4583	1.3555	1.2736	1.208	1.1559
0.65	2.3794	2.0130	1.7644	1.5856	1.4515	1.3479	1.2665	1.2021	1.1513	1.1116
0.6	2.0323	1.7680	1.5819	1.4447	1.3404	1.2596	1.1963	1.1469	1.1085	1.0791
0.55	1.7714	1.5778	1.4377	1.3329	1.2527	1.1907	1.1426	1.1056	1.0773	1.0559
0.5	1.5729	1.4302	1.3251	1.2458	1.1852	1.1386	1.1029	1.0757	1.0551	1.0398
Z_3										
$\delta_T \backslash \delta_A$	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
0.95	17.5607	8.3484	5.2059	3.7704	2.9784	2.4868	2.1593	1.9364	1.7783	1.6367
0.9	9.8712	5.9385	4.1782	3.2307	2.6544	2.2746	2.0123	1.8267	1.6812	1.5402
0.85	6.5133	4.5276	3.4562	2.8074	2.3817	2.0869	1.8754	1.7160	1.5800	1.4442
0.8	4.7347	3.6130	2.9231	2.4670	2.1490	1.9186	1.7455	1.606	1.4791	1.3541
0.75	3.6600	2.9769	2.5149	2.1878	1.9477	1.7658	1.6217	1.4986	1.3837	1.2740
0.7	2.9501	2.5120	2.1936	1.9553	1.7720	1.6267	1.5057	1.3983	1.2983	1.2064
0.65	2.4516	2.1600	1.9358	1.7598	1.6185	1.5015	1.4005	1.3093	1.2259	1.1521
0.6	2.0866	1.8872	1.7266	1.5953	1.4857	1.3919	1.309	1.2342	1.1673	1.1102
0.55	1.8123	1.6732	1.5568	1.4582	1.3734	1.2991	1.2329	1.1737	1.1220	1.0791
0.5	1.6034	1.5052	1.4203	1.3464	1.2815	1.2238	1.1724	1.127	1.0882	1.0567

Table S33. ARL_I values of the EWMA TBEA schemes for different statistics Z_i when the means of both A and T experience a shift based on $\lambda = 0.2$ when $\sigma_{0,A} = 1$ and $\sigma_{0,T} = 5$ with variables A following the normal distribution and T following the gamma distribution.

Z_1										
$\delta_T \backslash \delta_A$	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
0.95	30.846	14.2189	8.3908	5.7563	5.7563	3.469	2.8892	2.4760	2.1679	1.9306
0.9	20.4425	10.6921	6.8379	4.939	4.9390	3.1475	2.6618	2.3063	2.0365	1.8265
0.85	14.4379	8.3860	5.7285	4.3168	4.3168	2.8798	2.4662	2.1570	1.9191	1.7327
0.8	10.7615	6.8102	4.9104	3.8314	3.8314	2.6527	2.2955	2.0242	1.8137	1.6481
0.75	8.3870	5.6912	4.2903	3.4446	3.4446	2.4569	2.1447	1.9051	1.7185	1.5716
0.7	6.7790	4.8700	3.8091	3.1300	3.1300	2.2850	2.0097	1.7975	1.6323	1.5025
0.65	5.6445	4.2503	3.4280	2.8689	2.8689	2.1319	1.8877	1.6999	1.5542	1.4402
0.6	4.8154	3.7719	3.1205	2.6478	2.6478	1.9936	1.7767	1.6112	1.4835	1.384
0.55	4.1917	3.3956	2.8680	2.4564	2.4564	1.8672	1.6752	1.5304	1.4195	1.3336
0.5	3.7112	3.0950	2.6571	2.287	2.2870	1.7506	1.5821	1.4568	1.3617	1.2884
Z_2										
$\delta_T \backslash \delta_A$	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
0.95	72.0599	51.2581	37.8212	28.8483	22.6568	18.2549	15.0421	12.6425	10.813	9.3918
0.9	42.3498	31.5840	24.3482	19.3208	15.7207	13.0734	11.0812	9.5509	8.3535	7.4008
0.85	26.4080	20.5601	16.4749	13.5318	11.3534	9.7028	8.4263	7.4209	6.6158	5.9614
0.8	17.3727	14.0490	11.6431	9.8528	8.4882	7.4264	6.5850	5.9072	5.3532	4.8943
0.75	11.9822	10.0152	8.5453	7.4196	6.5391	5.8374	5.2692	4.8022	4.4135	4.086
0.7	8.6130	7.4074	6.4808	5.7528	5.1700	4.6957	4.3041	3.9766	3.6994	3.4625
0.65	6.4178	5.6569	5.0571	4.5752	4.1813	3.8547	3.5803	3.3472	3.1472	2.9738
0.6	4.9343	4.4422	4.0456	3.7205	3.4498	3.2216	3.027	2.8594	2.7136	2.5858
0.55	3.8990	3.5746	3.3081	3.0856	2.8974	2.7363	2.5971	2.4756	2.3688	2.2743
0.5	3.1559	2.9393	2.7581	2.6045	2.4726	2.3583	2.2583	2.1701	2.0917	2.0217
Z_3										
$\delta_T \backslash \delta_A$	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
0.95	84.8975	70.4614	57.2378	45.8108	36.4525	29.0552	23.3064	18.8723	15.4638	12.8442
0.9	46.0610	39.1209	32.8393	27.3954	22.8399	19.1070	16.0779	13.6309	11.6578	10.0663
0.85	27.0899	23.6441	20.5239	17.7813	15.4226	13.4192	11.7281	10.3050	9.1092	8.1044
0.8	17.2067	15.4265	13.7988	12.3391	11.0477	9.9141	8.9236	8.0603	7.3092	6.656
0.75	11.6886	10.7288	9.8382	9.0221	8.2808	7.6111	7.0084	6.4674	5.9827	5.5491
0.7	8.3901	7.8510	7.3426	6.8672	6.4252	6.0160	5.6386	5.2913	4.9725	4.6804
0.65	6.2918	5.9781	5.6776	5.3914	5.1201	4.8638	4.6225	4.3958	4.1834	3.9849
0.6	4.8839	4.6960	4.5136	4.3371	4.1671	4.0037	3.8472	3.6977	3.5552	3.4198
0.55	3.8969	3.7819	3.669	3.5584	3.4504	3.3451	3.2428	3.1437	3.0479	2.9555
0.5	3.1802	3.1089	3.0381	2.9681	2.8990	2.8309	2.7639	2.6982	2.634	2.5712

Table S34. ARL_I values of the EWMA TBEA schemes for different statistics Z_i when the means of both A and T experience a shift based on $\lambda = 0.5$ when $\sigma_{0,A} = 1$ and $\sigma_{0,T} = 1$ with variables A following the normal distribution and T following the gamma distribution.

Z_1										
$\delta_T \backslash \delta_A$	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
0.95	8.6715	3.2195	1.8907	1.3721	1.1325	1.0351	1.0066	1.0009	1.0001	1
0.9	4.8671	2.3774	1.5805	1.2288	1.0713	1.0160	1.0025	1.0003	1	1
0.85	3.2171	1.8896	1.3717	1.1325	1.0354	1.0067	1.0009	1.0001	1	1
0.8	2.3756	1.5797	1.2286	1.0715	1.0162	1.0026	1.0003	1	1	1
0.75	1.8883	1.3712	1.1326	1.0357	1.0069	1.0009	1.0001	1	1	1
0.7	1.5787	1.2284	1.0718	1.0165	1.0027	1.0003	1	1	1	1
0.65	1.3705	1.1327	1.0361	1.0071	1.0010	1.0001	1	1	1	1
0.6	1.2280	1.0722	1.0169	1.0029	1.0004	1	1	1	1	1
0.55	1.1328	1.0366	1.0075	1.0011	1.0001	1	1	1	1	1
0.5	1.0727	1.0175	1.0031	1.0004	1.0000	1	1	1	1	1
Z_2										
$\delta_T \backslash \delta_A$	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
0.95	9.1570	3.758	2.2817	1.6699	1.3473	1.1619	1.0636	1.0205	1.0054	1.0012
0.9	4.8279	2.6115	1.8080	1.4174	1.1999	1.0822	1.0280	1.0078	1.0018	1.0003
0.85	3.0890	1.9883	1.5027	1.2449	1.1049	1.0376	1.0111	1.0027	1.0005	1.0001
0.8	2.2318	1.6090	1.2989	1.1324	1.0499	1.0156	1.0040	1.0009	1.0002	1
0.75	1.7452	1.3642	1.1656	1.0652	1.0215	1.0059	1.0014	1.0003	1	1
0.7	1.4449	1.2056	1.0842	1.0293	1.0085	1.0021	1.0004	1.0001	1	1
0.65	1.2541	1.1075	1.0392	1.0121	1.0031	1.0007	1.0001	1	1	1
0.6	1.1360	1.0518	1.0168	1.0046	1.0011	1.0002	1.0000	1	1	1
0.55	1.0676	1.0231	1.0068	1.0017	1.0004	1.0001	1.0000	1	1	1
0.5	1.0313	1.0097	1.0026	1.0006	1.0001	1.0000	1	1	1	1
Z_3										
$\delta_T \backslash \delta_A$	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
0.95	8.9316	3.6443	2.1978	1.5950	1.2676	1.0894	1.0192	1.0024	1.0002	1
0.9	4.7619	2.6116	1.8013	1.3941	1.1614	1.0468	1.0086	1.0009	1.0001	1
0.85	3.0587	2.0163	1.5220	1.2433	1.0886	1.0223	1.0036	1.0003	1	1
0.8	2.2108	1.6384	1.3221	1.1375	1.0441	1.0097	1.0014	1.0001	1	1
0.75	1.7257	1.3860	1.1837	1.0705	1.0199	1.0038	1.0005	1	1	1
0.7	1.4250	1.2183	1.0955	1.0327	1.0082	1.0014	1.0002	1	1	1
0.65	1.2352	1.1132	1.0451	1.0139	1.0031	1.0005	1.0001	1	1	1
0.6	1.1204	1.0536	1.0195	1.0055	1.0011	1.0002	1	1	1	1
0.55	1.0567	1.0234	1.0078	1.0020	1.0004	1.0001	1	1	1	1
0.5	1.0248	1.0096	1.0030	1.0007	1.0001	1	1	1	1	1

Table S35. ARL_I values of the EWMA TBEA schemes for different statistics Z_i when the means of both A and T experience a shift based on $\lambda = 0.5$ when $\sigma_{0,A} = 1$ and $\sigma_{0,T} = 2$ with variables A following the normal distribution and T following the gamma distribution.

Z_1										
$\delta_T \backslash \delta_A$	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
0.95	16.9313	6.3323	3.4156	2.2819	1.7255	1.4121	1.2253	1.1155	1.0547	1.0238
0.9	9.7949	4.4864	2.7331	1.9584	1.5467	1.3061	1.1628	1.0806	1.0368	1.0155
0.85	6.3286	3.4052	2.2752	1.7215	1.4099	1.2246	1.1158	1.0554	1.0245	1.0101
0.8	4.4744	2.7237	1.9525	1.5433	1.3045	1.1626	1.0813	1.0377	1.0163	1.0065
0.75	3.3925	2.2669	1.7163	1.4071	1.2237	1.1162	1.0564	1.0255	1.0108	1.0043
0.7	2.7120	1.9451	1.5388	1.3024	1.1623	1.0821	1.0388	1.0172	1.0072	1.0028
0.65	2.2563	1.7096	1.4033	1.2223	1.1165	1.0575	1.0267	1.0117	1.0048	1.0019
0.6	1.9352	1.5327	1.2994	1.1618	1.0830	1.0402	1.0185	1.0080	1.0033	1.0013
0.55	1.7003	1.3980	1.2204	1.1169	1.0590	1.0283	1.0129	1.0057	1.0024	1.0010
0.5	1.5240	1.2951	1.1609	1.0841	1.0420	1.0201	1.0093	1.0041	1.0018	1.0007
Z_2										
$\delta_T \backslash \delta_A$	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
0.95	24.731	10.9034	6.0556	3.9873	2.9436	2.3444	1.9657	1.7086	1.5246	1.388
0.9	12.7038	6.7137	4.2645	3.074	2.4103	2.0003	1.7270	1.5343	1.3932	1.2874
0.85	7.5058	4.5873	3.2214	2.4831	2.0379	1.7467	1.5447	1.3987	1.2904	1.2091
0.8	4.9637	3.3883	2.5636	2.0786	1.7678	1.5557	1.4045	1.2935	1.211	1.1495
0.75	3.5771	2.6523	2.1227	1.7903	1.5673	1.4105	1.2968	1.213	1.1509	1.1053
0.7	2.7498	2.1700	1.8140	1.5793	1.4166	1.3001	1.215	1.1524	1.1066	1.0733
0.65	2.2205	1.8387	1.5916	1.4229	1.3034	1.217	1.1539	1.1079	1.0746	1.0508
0.6	1.8642	1.6040	1.4290	1.3066	1.219	1.1554	1.1092	1.0759	1.052	1.0351
0.55	1.6160	1.4348	1.3096	1.2208	1.1569	1.1106	1.0773	1.0534	1.0364	1.0245
0.5	1.4398	1.3121	1.2224	1.1583	1.1121	1.0788	1.0548	1.0378	1.0258	1.0174
Z_3										
$\delta_T \backslash \delta_A$	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
0.95	29.2275	13.3583	7.0495	4.3428	3.0375	2.3305	1.9069	1.6268	1.4183	1.2498
0.9	14.2252	7.8684	4.8859	3.3821	2.5535	2.0565	1.7341	1.5061	1.3294	1.1887
0.85	8.1066	5.2127	3.6456	2.7458	2.1946	1.8354	1.5858	1.3997	1.2527	1.1391
0.8	5.2374	3.7667	2.8705	2.3001	1.9199	1.6544	1.4593	1.3084	1.1892	1.1003
0.75	3.7179	2.9004	2.3529	1.9749	1.7051	1.5059	1.3531	1.2325	1.1385	1.0711
0.7	2.8293	2.3417	1.9899	1.7307	1.5355	1.3848	1.2659	1.1716	1.0996	1.0497
0.65	2.2684	1.9615	1.7267	1.5447	1.4017	1.2878	1.1965	1.1244	1.0706	1.0346
0.6	1.8942	1.6932	1.5322	1.4025	1.2973	1.2118	1.1429	1.0891	1.0498	1.0240
0.55	1.6352	1.4998	1.3876	1.2945	1.2173	1.154	1.1029	1.0634	1.0351	1.0168
0.5	1.4522	1.3594	1.2804	1.2136	1.1575	1.1111	1.0738	1.0453	1.0250	1.0120

Table S36. ARL_I values of the EWMA TBEA schemes for different statistics Z_i when the means of both A and T experience a shift based on $\lambda = 0.5$ when $\sigma_{0,A} = 1$ and $\sigma_{0,T} = 5$ with variables A following the normal distribution and T following the gamma distribution.

Z_1										
$\delta_T \backslash \delta_A$	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
0.95	32.2059	13.9743	7.7706	5.0746	5.0746	2.8731	2.3583	2.0103	1.7642	1.5844
0.9	21.4001	10.3560	6.2108	4.2718	4.2718	2.5752	2.1564	1.8673	1.6598	1.5066
0.85	14.9157	7.9534	5.0918	3.6614	3.6614	2.3311	1.9873	1.7457	1.5701	1.4394
0.8	10.8634	6.3032	4.2668	3.1872	3.1872	2.1287	1.8443	1.6416	1.4928	1.3811
0.75	8.2307	5.1333	3.6429	2.8114	2.8114	1.9590	1.7227	1.5522	1.4259	1.3307
0.7	6.4558	4.2787	3.1597	2.5082	2.5082	1.8156	1.6186	1.4751	1.3681	1.2870
0.65	5.2172	3.6369	2.7774	2.2596	2.2596	1.6934	1.5292	1.4086	1.3181	1.2491
0.6	4.3244	3.1421	2.4687	2.0529	2.0529	1.5888	1.4521	1.3510	1.2747	1.2163
0.55	3.6612	2.7511	2.2150	1.8789	1.8789	1.4987	1.3854	1.3011	1.2372	1.1880
0.5	3.1545	2.4350	2.0029	1.7309	1.7309	1.4209	1.3276	1.2578	1.2045	1.1633
Z_2										
$\delta_T \backslash \delta_A$	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
0.95	78.6884	58.2388	44.1873	34.2974	27.1870	21.9745	18.0834	15.1287	12.8484	11.0617
0.9	46.5204	35.5558	27.8313	22.2622	18.1628	15.0872	12.7386	10.9152	9.4776	8.3278
0.85	28.8011	22.7241	18.3358	15.0953	12.6536	10.7796	9.3166	8.1565	7.2232	6.4624
0.8	18.6639	15.1848	12.6105	10.6641	9.1637	7.9865	7.0481	6.2891	5.6669	5.1507
0.75	12.6398	10.5847	9.0272	7.8224	6.8731	6.1128	5.4947	4.9855	4.5608	4.2028
0.7	8.9227	7.6721	6.7024	5.9357	5.3190	4.8155	4.3988	4.0497	3.754	3.5011
0.65	6.5435	5.7615	5.1418	4.6417	4.2319	3.8913	3.6048	3.3611	3.1518	2.9704
0.6	4.9666	4.4658	4.0608	3.7279	3.4502	3.2158	3.0156	2.8431	2.6931	2.5615
0.55	3.8871	3.5600	3.2905	3.0651	2.8743	2.7108	2.5694	2.446	2.3375	2.2414
0.5	3.1261	2.9092	2.7275	2.5733	2.4408	2.3259	2.2253	2.1365	2.0577	1.9873
Z_3										
$\delta_T \backslash \delta_A$	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
0.95	100.2309	92.0414	83.9490	76.0251	68.3799	61.1438	54.4269	48.2827	42.7094	37.6783
0.9	57.3210	53.0109	48.7566	44.5974	40.5865	36.7806	33.2239	29.9364	26.9169	24.1547
0.85	34.4045	32.0701	29.7679	27.5182	25.3460	23.2760	21.3262	19.5046	17.8112	16.2423
0.8	21.6726	20.3736	19.0926	17.8398	16.6267	15.4639	14.3593	13.3164	12.3356	11.4158
0.75	14.3126	13.5710	12.8391	12.1216	11.4239	10.7507	10.1056	9.4902	8.9051	8.3502
0.7	9.8862	9.4525	9.0235	8.6016	8.1893	7.7886	7.4013	7.0283	6.67	6.3266
0.65	7.1184	6.8589	6.6016	6.3476	6.0978	5.8534	5.6152	5.3838	5.1594	4.9422
0.6	5.3215	5.1632	5.0056	4.8494	4.6949	4.5427	4.3932	4.2468	4.1036	3.9638
0.55	4.1132	4.0149	3.9167	3.8189	3.7217	3.6254	3.5301	3.436	3.3434	3.2522
0.5	3.2738	3.2120	3.1500	3.0881	3.0261	2.9644	2.903	2.842	2.7815	2.7215

Table S37. ARL_I values of the EWMA TBEA schemes for different statistics Z_i when the means of both A and T experience a shift based on $\lambda = 0.8$ when $\sigma_{0,A} = 1$ and $\sigma_{0,T} = 1$ with variables A following the normal distribution and T following the gamma distribution.

Z_1										
$\delta_T \backslash \delta_A$	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
0.95	11.681	3.7539	1.9258	1.3258	1.1002	1.0236	1.004	1.0005	1	1
0.9	6.1716	2.5658	1.5524	1.1855	1.0507	1.0102	1.0015	1.0001	1	1
0.85	3.7506	1.9240	1.3254	1.1003	1.0238	1.0041	1.0005	1.0000	1	1
0.8	2.5627	1.5512	1.1854	1.0510	1.0104	1.0015	1.0002	1	1	1
0.75	1.9217	1.3248	1.1006	1.0241	1.0042	1.0005	1	1	1	1
0.7	1.5498	1.1853	1.0513	1.0107	1.0016	1.0002	1	1	1	1
0.65	1.3241	1.1008	1.0245	1.0044	1.0006	1.0001	1	1	1	1
0.6	1.1852	1.0518	1.0110	1.0017	1.0002	1	1	1	1	1
0.55	1.1011	1.0251	1.0047	1.0007	1.0001	1	1	1	1	1
0.5	1.0524	1.0115	1.0019	1.0002	1.0000	1	1	1	1	1
Z_2										
$\delta_T \backslash \delta_A$	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
0.95	12.8504	4.6553	2.4339	1.6259	1.2716	1.1071	1.036	1.01	1.0022	1.0004
0.9	6.0999	2.8718	1.7947	1.3465	1.1409	1.0498	1.0147	1.0035	1.0007	1.0001
0.85	3.4875	2.0160	1.4409	1.1836	1.0678	1.0211	1.0054	1.0012	1.0002	1
0.8	2.3116	1.5610	1.2371	1.0911	1.0300	1.0082	1.0019	1.0003	1.0001	1
0.75	1.7159	1.3046	1.1207	1.0418	1.0122	1.0029	1.0006	1.0001	1	1
0.7	1.3899	1.1584	1.0573	1.0177	1.0046	1.0010	1.0002	1	1	1
0.65	1.2059	1.0774	1.0252	1.0069	1.0016	1.0003	1.0001	1	1	1
0.6	1.1033	1.0354	1.0103	1.0026	1.0005	1.0001	1	1	1	1
0.55	1.0488	1.0151	1.0040	1.0009	1.0002	1	1	1	1	1
0.5	1.0217	1.0062	1.0015	1.0003	1.0001	1	1	1	1	1
Z_3										
$\delta_T \backslash \delta_A$	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
0.95	12.7432	4.5050	2.2846	1.4998	1.1734	1.0447	1.0073	1.0007	1	1
0.9	6.0166	2.8502	1.7513	1.2938	1.0947	1.0216	1.0031	1.0003	1	1
0.85	3.4162	2.0203	1.4325	1.1630	1.0476	1.0095	1.0012	1.0001	1	1
0.8	2.2529	1.5668	1.2388	1.0839	1.0219	1.0039	1.0004	1	1	1
0.75	1.6686	1.3071	1.1237	1.0397	1.0092	1.0014	1.0001	1	1	1
0.7	1.3530	1.1580	1.0593	1.0172	1.0036	1.0005	1	1	1	1
0.65	1.1788	1.0758	1.0262	1.0069	1.0013	1.0002	1	1	1	1
0.6	1.0851	1.0338	1.0107	1.0026	1.0005	1.0001	1	1	1	1
0.55	1.0379	1.0140	1.0042	1.0009	1.0002	1	1	1	1	1
0.5	1.0159	1.0056	1.0016	1.0003	1.0001	1	1	1	1	1

Table S38. ARL_I values of the EWMA TBEA schemes for different statistics Z_i when the means of both A and T experience a shift based on $\lambda = 0.8$ when $\sigma_{0,A} = 1$ and $\sigma_{0,T} = 2$ with variables A following the normal distribution and T following the gamma distribution.

Z_1										
$\delta_T \backslash \delta_A$	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
0.95	21.9268	7.8600	3.8427	2.3591	1.6959	1.3613	1.1829	1.0879	1.0395	1.0164
0.9	12.6450	5.3129	2.9349	1.9643	1.5002	1.2578	1.1279	1.0598	1.0261	1.0106
0.85	7.8967	3.8309	2.3489	1.6903	1.3590	1.1825	1.0885	1.0403	1.0171	1.0068
0.8	5.3111	2.9213	1.9556	1.4959	1.2564	1.1281	1.0607	1.0270	1.0112	1.0044
0.75	3.8168	2.3362	1.6833	1.3560	1.1820	1.0892	1.0413	1.0180	1.0074	1.0028
0.7	2.9045	1.9445	1.4905	1.2546	1.1283	1.0617	1.0281	1.0121	1.0049	1.0019
0.65	2.3200	1.6741	1.3521	1.1813	1.0900	1.0426	1.0191	1.0081	1.0033	1.0013
0.6	1.9301	1.4832	1.2521	1.1285	1.0630	1.0295	1.0132	1.0056	1.0023	1.0009
0.55	1.6617	1.3466	1.1801	1.0910	1.0442	1.0206	1.0092	1.0040	1.0016	1.0007
0.5	1.4729	1.2484	1.1285	1.0646	1.0313	1.0147	1.0066	1.0029	1.0012	1.0005
Z_2										
$\delta_T \backslash \delta_A$	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
0.95	32.5828	15.2786	8.2845	5.1047	3.4966	2.6048	2.071	1.7314	1.5053	1.3498
0.9	16.7430	8.8742	5.3663	3.6228	2.6702	2.1069	1.7523	1.518	1.3581	1.2465
0.85	9.5311	5.6515	3.7578	2.7390	2.1442	1.7737	1.5310	1.3664	1.2522	1.1721
0.8	5.9629	3.9021	2.8111	2.1828	1.7955	1.5441	1.3747	1.2578	1.1762	1.119
0.75	4.0563	2.8866	2.2223	1.8176	1.5572	1.3830	1.2635	1.1803	1.1222	1.0817
0.7	2.9653	2.2627	1.8397	1.5702	1.3913	1.2691	1.1844	1.1253	1.0842	1.0558
0.65	2.3035	1.8616	1.5829	1.3992	1.2746	1.1885	1.1285	1.0868	1.0579	1.0381
0.6	1.8828	1.5949	1.4068	1.2798	1.1925	1.1317	1.0894	1.0601	1.0399	1.0261
0.55	1.6060	1.4137	1.2847	1.1962	1.1349	1.0921	1.0624	1.0418	1.0277	1.0182
0.5	1.4195	1.2889	1.1997	1.138	1.0949	1.0648	1.0439	1.0295	1.0196	1.0129
Z_3										
$\delta_T \backslash \delta_A$	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
0.95	40.3212	21.1458	11.4696	6.6024	4.1058	2.7765	2.0335	1.5953	1.3249	1.1571
0.9	19.5555	11.4810	7.0005	4.5065	3.0912	2.2630	1.7602	1.4435	1.2401	1.1131
0.85	10.6424	6.9201	4.6668	3.2948	2.4455	1.9075	1.5581	1.3269	1.1746	1.0801
0.8	6.4376	4.5729	3.3531	2.5505	2.0159	1.6546	1.4073	1.2378	1.1250	1.0558
0.75	4.2737	3.2678	2.5641	2.0698	1.7203	1.4717	1.2948	1.1709	1.0882	1.0385
0.7	3.0709	2.4926	2.0645	1.7474	1.5123	1.3384	1.2113	1.1213	1.0616	1.0264
0.65	2.3572	2.0068	1.7352	1.5253	1.3639	1.2413	1.1501	1.0854	1.0428	1.0181
0.6	1.9110	1.6897	1.5118	1.3699	1.2578	1.1711	1.106	1.0598	1.0297	1.0125
0.55	1.6210	1.4769	1.3578	1.2605	1.1822	1.1209	1.0747	1.042	1.0208	1.0088
0.5	1.4275	1.3317	1.2508	1.1836	1.1288	1.0856	1.0529	1.0298	1.0148	1.0063

Table S39. ARL_I values of the EWMA TBEA schemes for different statistics Z_i when the means of both A and T experience a shift based on $\lambda = 0.8$ when $\sigma_{0,A} = 1$ and $\sigma_{0,T} = 5$ with variables A following the normal distribution and T following the gamma distribution.

Z_1										
$\delta_T \backslash \delta_A$	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
0.95	34.7466	14.6201	7.8652	4.9902	4.9902	2.723	2.2164	1.8833	1.6541	1.4908
0.9	23.5765	10.8186	6.2230	4.153	4.1530	2.4250	2.0198	1.7481	1.5581	1.4213
0.85	16.5034	8.2240	5.0330	3.5174	3.5174	2.1842	1.8579	1.6350	1.4772	1.3622
0.8	11.9113	6.4129	4.1540	3.0273	3.0273	1.9879	1.7236	1.5402	1.4086	1.312
0.75	8.8547	5.1206	3.4928	2.6439	2.6439	1.8266	1.6116	1.4602	1.3505	1.2692
0.7	6.7692	4.1788	2.9869	2.3401	2.3401	1.6931	1.5177	1.3927	1.3011	1.2327
0.65	5.3114	3.4784	2.5936	2.0964	2.0964	1.582	1.4388	1.3354	1.2591	1.2015
0.6	4.2679	2.9476	2.2835	1.8988	1.8988	1.4890	1.3721	1.2868	1.2232	1.175
0.55	3.5038	2.5381	2.0356	1.7371	1.7371	1.4107	1.3155	1.2454	1.1926	1.1522
0.5	2.9320	2.2168	1.8351	1.6037	1.6037	1.3445	1.2675	1.21	1.1664	1.1327
Z_2										
$\delta_T \backslash \delta_A$	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
0.95	82.9527	63.2371	49.3726	39.3533	31.9408	26.3433	22.0396	18.6772	16.0122	13.8725
0.9	49.8920	39.1189	31.3488	25.601	21.2556	17.9074	15.2839	13.1975	11.516	10.1444
0.85	31.2258	25.1504	20.6622	17.2679	14.6486	12.5915	10.9505	9.6234	8.5368	7.6372
0.8	20.3186	16.7887	14.1212	12.0613	10.4407	9.1447	8.0933	7.2295	6.5117	5.9091
0.75	13.7303	11.6213	9.9932	8.7110	7.6836	6.8481	6.1595	5.5853	5.1015	4.69
0.7	9.6218	8.3288	7.3106	6.4937	5.8280	5.2779	4.8178	4.4289	4.097	3.8112
0.65	6.9803	6.1690	5.5180	4.9868	4.5469	4.178	3.8653	3.5975	3.3663	3.1649
0.6	5.2322	4.7126	4.2885	3.9367	3.6412	3.39	3.1743	2.9875	2.8244	2.6809
0.55	4.0433	3.7049	3.4241	3.1879	2.9867	2.8135	2.6632	2.5315	2.4153	2.3122
0.5	3.2139	2.9906	2.8026	2.6423	2.5041	2.3837	2.2781	2.1847	2.1016	2.0272
Z_3										
$\delta_T \backslash \delta_A$	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
0.95	104.9557	98.7798	92.7777	86.9481	81.2963	75.8355	70.5861	65.5712	60.8081	56.3027
0.9	61.3334	58.0686	54.8806	51.7703	48.7419	45.8027	42.9633	40.2348	37.6256	35.1388
0.85	37.3848	35.6004	33.8505	32.1363	30.4603	28.8267	27.2409	25.7082	24.2329	22.8167
0.8	23.7452	22.7391	21.7487	20.7747	19.8188	18.8831	17.9703	17.0833	16.2242	15.3939
0.75	15.6969	15.1132	14.5365	13.9674	13.4066	12.8555	12.3154	11.7877	11.2735	10.7735
0.7	10.7838	10.4361	10.0915	9.7503	9.4128	9.0798	8.752	8.4301	8.1147	7.8062
0.65	7.6854	7.4735	7.2627	7.0533	6.8456	6.6398	6.4363	6.2355	6.0377	5.8432
0.6	5.6703	5.5384	5.4069	5.2758	5.1453	5.0156	4.8868	4.7591	4.6328	4.5079
0.55	4.3211	4.2376	4.1542	4.0707	3.9874	3.9043	3.8214	3.739	3.657	3.5756
0.5	3.3929	3.3393	3.2857	3.2319	3.1780	3.1241	3.0702	3.0163	2.9625	2.9089

Table S40. ARL_I values of the EWMA TBEA schemes for different statistics Z_i when the means of both A and T experience a shift based on $\lambda = 1$ when $\sigma_{0,A} = 1$ and $\sigma_{0,T} = 1$ with variables A following the normal distribution and T following the gamma distribution.

Z_1										
$\delta_T \backslash \delta_A$	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
0.95	14.0369	4.3532	2.0410	1.3319	1.0939	1.0209	1.0034	1.0004	1	1
0.9	7.3941	2.8423	1.5908	1.1807	1.0461	1.0089	1.0012	1.0001	1	1
0.85	4.3506	2.0386	1.3313	1.0941	1.0211	1.0035	1.0004	1.0000	1	1
0.8	2.8386	1.5893	1.1806	1.0463	1.0090	1.0013	1.0001	1	1	1
0.75	2.0356	1.3306	1.0943	1.0214	1.0036	1.0004	1	1	1	1
0.7	1.5874	1.1805	1.0467	1.0093	1.0014	1.0001	1	1	1	1
0.65	1.3297	1.0946	1.0218	1.0038	1.0005	1	1	1	1	1
0.6	1.1803	1.0472	1.0096	1.0015	1.0002	1	1	1	1	1
0.55	1.0950	1.0223	1.004	1.0005	1.0001	1	1	1	1	1
0.5	1.0478	1.0101	1.0016	1.0002	1.0000	1	1	1	1	1
Z_2										
$\delta_T \backslash \delta_A$	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
0.95	15.5665	5.591	2.7043	1.6789	1.2670	1.0964	1.0302	1.0079	1.0017	1.0003
0.9	7.2434	3.2301	1.8781	1.3495	1.1304	1.0429	1.0119	1.0027	1.0005	1.0001
0.85	3.9424	2.1365	1.4548	1.1743	1.0599	1.0176	1.0043	1.0009	1.0001	1
0.8	2.4749	1.5892	1.2305	1.0825	1.0256	1.0067	1.0014	1.0003	1	1
0.75	1.7616	1.3023	1.1119	1.0365	1.0101	1.0023	1.0005	1.0001	1	1
0.7	1.3937	1.1500	1.0513	1.0151	1.0037	1.0008	1.0001	1	1	1
0.65	1.1988	1.0708	1.0220	1.0058	1.0013	1.0002	1	1	1	1
0.6	1.0963	1.0315	1.0089	1.0021	1.0004	1.0001	1	1	1	1
0.55	1.0443	1.0133	1.0034	1.0008	1.0001	1	1	1	1	1
0.5	1.0194	1.0053	1.0013	1.0003	1	1	1	1	1	1
Z_3										
$\delta_T \backslash \delta_A$	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
0.95	15.7985	5.5259	2.5361	1.5242	1.1580	1.0361	1.0054	1.0005	1	1
0.9	7.2519	3.2403	1.8277	1.2862	1.0818	1.0169	1.0022	1.0002	1	1
0.85	3.8820	2.1478	1.4414	1.1492	1.0395	1.0072	1.0008	1.0001	1	1
0.8	2.4063	1.5930	1.2281	1.0731	1.0176	1.0029	1.0003	1	1	1
0.75	1.7034	1.3011	1.1121	1.0333	1.0072	1.0011	1.0001	1	1	1
0.7	1.3499	1.1469	1.0516	1.0141	1.0028	1.0004	1	1	1	1
0.65	1.1687	1.0677	1.0222	1.0055	1.0010	1.0001	1	1	1	1
0.6	1.0774	1.0293	1.0089	1.0021	1.0003	1	1	1	1	1
0.55	1.0336	1.0120	1.0034	1.0007	1.0001	1	1	1	1	1
0.5	1.0139	1.0047	1.0013	1.0003	1.0000	1	1	1	1	1

Table S41. ARL_I values of the EWMA TBEA schemes for different statistics Z_i when the means of both A and T experience a shift based on $\lambda = 1$ when $\sigma_{0,A} = 1$ and $\sigma_{0,T} = 2$ with variables A following the normal distribution and T following the gamma distribution.

Z_1										
$\delta_T \backslash \delta_A$	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
0.95	25.4521	9.2521	4.3504	2.5283	1.7406	1.3649	1.1767	1.0821	1.0359	1.0147
0.9	14.9534	6.1686	3.2325	2.0545	1.5178	1.2543	1.1214	1.0552	1.0236	1.0094
0.85	9.3407	4.3430	2.5161	1.7336	1.3622	1.1764	1.0827	1.0368	1.0154	1.0060
0.8	6.1879	3.2183	2.0437	1.5126	1.2528	1.1217	1.0561	1.0244	1.0100	1.0039
0.75	4.3349	2.5010	1.7249	1.3587	1.1759	1.0835	1.0378	1.0162	1.0066	1.0025
0.7	3.2008	2.0300	1.5060	1.2508	1.1220	1.0571	1.0255	1.0108	1.0043	1.0017
0.65	2.4817	1.7136	1.3541	1.1751	1.0844	1.0391	1.0173	1.0073	1.0029	1.0011
0.6	2.0122	1.4973	1.2480	1.1222	1.0585	1.0269	1.0119	1.0050	1.0020	1.0008
0.55	1.6984	1.3478	1.1739	1.0855	1.0408	1.0187	1.0083	1.0035	1.0015	1.0006
0.5	1.4851	1.2440	1.1223	1.0602	1.0288	1.0133	1.0060	1.0026	1.0011	1.0005
Z_2										
$\delta_T \backslash \delta_A$	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
0.95	36.3737	17.8898	9.8911	6.0520	4.0420	2.9111	2.2365	1.8151	1.5426	1.3621
0.9	18.9854	10.3235	6.2422	4.1343	2.9601	2.2648	1.8327	1.5543	1.3703	1.2467
0.85	10.8036	6.4465	4.2307	3.0101	2.2932	1.8502	1.5659	1.3784	1.2526	1.1675
0.8	6.6680	4.3318	3.0612	2.3215	1.8675	1.5772	1.3864	1.2585	1.172	1.1133
0.75	4.4383	3.1132	2.3497	1.8842	1.5882	1.3941	1.2642	1.1764	1.1168	1.0764
0.7	3.1661	2.3773	1.9004	1.5987	1.4015	1.2698	1.1808	1.1203	1.0792	1.0515
0.65	2.4043	1.9157	1.6085	1.4085	1.2752	1.1851	1.1238	1.082	1.0538	1.0349
0.6	1.9297	1.6173	1.4149	1.2802	1.1892	1.1273	1.085	1.0562	1.0368	1.0238
0.55	1.6248	1.4204	1.2847	1.1932	1.1307	1.0879	1.0587	1.0389	1.0255	1.0165
0.5	1.4247	1.2886	1.1967	1.134	1.0909	1.0614	1.0411	1.0273	1.018	1.0117
Z_3										
$\delta_T \backslash \delta_A$	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
0.95	45.6928	25.9777	14.9992	8.8644	5.4147	3.4587	2.3406	1.7000	1.3381	1.1434
0.9	22.5551	13.9898	8.8168	5.6853	3.7828	2.6228	1.9152	1.4875	1.2367	1.0995
0.85	12.2656	8.2293	5.6103	3.9100	2.8058	2.0899	1.6294	1.3393	1.1647	1.0684
0.8	7.3025	5.2554	3.8415	2.8668	2.1972	1.7407	1.4344	1.2355	1.1138	1.0466
0.75	4.7273	3.6192	2.8120	2.2268	1.8057	1.5070	1.3001	1.1628	1.0781	1.0316
0.7	3.3027	2.6683	2.1852	1.8201	1.5477	1.3484	1.2071	1.1122	1.0535	1.0214
0.65	2.4705	2.0899	1.7894	1.5548	1.3746	1.2399	1.1429	1.0772	1.0366	1.0145
0.6	1.9622	1.7251	1.5325	1.3783	1.2574	1.1655	1.0987	1.0533	1.0252	1.0100
0.55	1.6408	1.4888	1.3626	1.2596	1.1776	1.1146	1.0685	1.037	1.0175	1.0070
0.5	1.4324	1.3329	1.2489	1.1794	1.1234	1.0799	1.048	1.0261	1.0124	1.0050

Table S42. ARL_I values of the EWMA TBEA schemes for different statistics Z_i when the means of both A and T experience a shift based on $\lambda = 1$ when $\sigma_{0,A} = 1$ and $\sigma_{0,T} = 5$ with variables A following the normal distribution and T following the gamma distribution.

Z_1										
$\delta_T \backslash \delta_A$	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
0.95	38.3722	16.1689	8.5530	5.3079	5.3079	2.7883	2.2406	1.8871	1.6481	1.4807
0.9	26.6422	12.0201	6.7346	4.3808	4.3808	2.4654	2.0310	1.7453	1.5492	1.4102
0.85	18.9139	9.1224	5.4024	3.6752	3.6752	2.2063	1.8598	1.6279	1.4665	1.3508
0.8	13.7245	7.0643	4.4125	3.1320	3.132	1.9968	1.7192	1.5304	1.3972	1.3007
0.75	10.1752	5.5790	3.6669	2.7091	2.7091	1.8264	1.6032	1.449	1.3389	1.2584
0.7	7.7043	4.4907	3.0983	2.3766	2.3766	1.6869	1.507	1.3809	1.2899	1.2226
0.65	5.9545	3.6817	2.6596	2.1127	2.1127	1.5721	1.4269	1.3238	1.2485	1.1923
0.6	4.6950	3.0722	2.3174	1.9016	1.9016	1.4772	1.36	1.2757	1.2135	1.1667
0.55	3.7743	2.6071	2.0478	1.7312	1.7312	1.3983	1.3039	1.2352	1.1839	1.1448
0.5	3.0913	2.2479	1.8334	1.5928	1.5928	1.3324	1.2567	1.2008	1.1586	1.1262
Z_2										
$\delta_T \backslash \delta_A$	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
0.95	84.4698	65.0394	51.2896	41.2834	33.8237	28.1447	23.741	20.2703	17.495	15.2469
0.9	51.1586	40.4804	32.7298	26.9565	22.5592	19.1444	16.4472	14.2848	12.5277	11.083
0.85	32.1875	26.1336	21.6331	18.2065	15.5435	13.4366	11.7435	10.3641	9.2264	8.2778
0.8	21.0133	17.4791	14.7922	12.7040	11.0502	9.7188	8.6316	7.7324	6.9803	6.345
0.75	14.2164	12.0963	10.4503	9.1464	8.0954	7.2354	6.5225	5.9247	5.4181	4.985
0.7	9.9538	8.6499	7.6177	6.7852	6.1032	5.5367	5.0605	4.656	4.3092	4.0092
0.65	7.2022	6.3822	5.7211	5.1792	4.7284	4.3488	4.0256	3.7477	3.5067	3.2962
0.6	5.3770	4.8512	4.4202	4.0615	3.7589	3.5008	3.2784	3.0851	2.9158	2.7665
0.55	4.1353	3.7927	3.5075	3.2669	3.0612	2.8838	2.7292	2.5935	2.4735	2.3668
0.5	3.2703	3.0444	2.8538	2.6907	2.5498	2.4269	2.3188	2.223	2.1375	2.0609
Z_3										
$\delta_T \backslash \delta_A$	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
0.95	106.4822	100.9196	95.5717	90.4334	85.4995	80.7648	76.2243	71.8730	67.7058	63.718
0.9	62.6961	59.7614	56.9239	54.1818	51.5334	48.9767	46.5101	44.1316	41.8396	39.6322
0.85	38.4435	36.8412	35.2838	33.7706	32.3010	30.8744	29.4902	28.1478	26.8466	25.5859
0.8	24.5159	23.6125	22.7300	21.8682	21.0269	20.2059	19.4051	18.6243	17.8634	17.122
0.75	16.2372	15.7123	15.1971	14.6916	14.1958	13.7097	13.2331	12.7661	12.3087	11.8607
0.7	11.1523	10.8388	10.5297	10.2252	9.9251	9.6296	9.3385	9.052	8.77	8.4925
0.65	7.931	7.7390	7.5490	7.3610	7.1750	6.9911	6.8092	6.6293	6.4515	6.2757
0.6	5.8301	5.7100	5.5906	5.4721	5.3544	5.2375	5.1215	5.0063	4.892	4.7785
0.55	4.4223	4.3457	4.2694	4.1933	4.1175	4.042	3.9667	3.8917	3.8171	3.7427
0.5	3.4548	3.4053	3.3558	3.3063	3.2568	3.2074	3.158	3.1086	3.0592	3.0099